Multivariate Data Analysis

Part I: Distributions and Inference (Ch. 1–6)

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Chapter 1. Multivariate Data and the Geometry of the Sample

Data matrices, descriptive statistics, projections, sphering, Mahalanobis distance, eigendecomposition

1 Multivariate Data and the Goals of Analysis

Multivariate (multidimensional) data are measurements of two or more variables on each individual or experimental unit. The goals of analysis fall into two groups:

(i) **Inference about an unknown population**: understand and infer features of the population from which the data were drawn.

(ii) **Prediction of future observations**: predict the value or category of new observations.

In either case one starts with exploratory data analysis (EDA) to find structure, outliers and relationships among variables. Numerical summaries are descriptive statistics (mean vector, covariance matrix, correlation matrix); graphical summaries are scatterplot matrices, principal component plots, heatmaps, correlation plots and trellis graphs.

Example 1.1: two running examples

Penguins data. For $n = 333$ penguins from the Palmer archipelago there are 4 numerical variables (bill length, bill depth, flipper length, body mass), 3 categorical variables (species, island, sex) and year. Only the numerical variables go into the data matrix $\mathbf{X}$; categorical variables are kept as separate vectors. This is the typical case $n > p$.
Golub gene expression data (1999). For 72 leukemia patients the expression levels of 7129 genes were measured, together with a categorical label ALL/AML. This is high-dimensional, low-sample-size data ($p = 7129 \gg n = 72$). Scientific question: can the leukemia type be predicted from gene expression alone?

2 The Data Matrix and Descriptive Statistics

Motivation. Multivariate data are an $n \times p$ matrix, and almost every method in the course is a statement about this matrix. Writing the sample mean and covariance as matrix products lets us use linear algebra instead of indices.

2.1 Data matrix

Measurements of p variables on n items form an $n \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = [\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(p)}].$$

The ith row $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the **observation vector** of item i; the jth column $\mathbf{x}_{(j)} = (x_{1j}, \ldots, x_{nj})^\top \in \mathbb{R}^n$ is the **variable vector** of variable j. All variables in $\mathbf{X}$ are taken to be numerical.

2.2 Sample mean, covariance and correlation

Definition 1.2: descriptive statistics

$$\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i, \qquad \mathbf{S} = (s_{jk}), \quad s_{jk} = \frac{1}{n-1}\sum_{i=1}^{n}(x_{ij}-\bar{x}_j)(x_{ik}-\bar{x}_k),$$

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^\top, \qquad \mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}, \qquad \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2},$$

with $\mathbf{D} = \operatorname{diag}(s_{11},\dots,s_{pp})$. $\mathbf{R}$ is defined only when all $s_{jj} > 0$. One also uses $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$ (divisor n).

Remark 1.3: a Gaussian with covariance S always exists

$\mathbf{S}$ is symmetric and positive semidefinite, so a symmetric square root $\mathbf{S}^{1/2}$ exists (§5). With $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{S}^{1/2}\mathbf{Z}$, $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$ and $\mathrm{Cov}\mathbf{X} = \mathbf{S}^{1/2}\mathbf{I}_p\mathbf{S}^{1/2} = \mathbf{S}$.

2.3 The centering matrix

Let $\mathbf{1}_n = (1,\dots,1)^\top$, $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n^\top$ and $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

Theorem 1.4: matrix forms of $\bar{\mathbf{x}}$ and S

$$\bar{\mathbf{x}} = \tfrac{1}{n}\mathbf{X}^\top\mathbf{1}_n, \qquad \mathbf{X}_C := \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{H}_n\mathbf{X}, \qquad \mathbf{S} = \tfrac{1}{n-1}\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = \tfrac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C,$$

and $\mathbf{H}_n^\top = \mathbf{H}_n$, $\mathbf{H}_n^2 = \mathbf{H}_n$, $\operatorname{rank}(\mathbf{H}_n) = n-1$.

Proof. The jth entry of $\mathbf{X}^\top\mathbf{1}_n$ is $\sum_i x_{ij}$, giving the first identity. Hence $\mathbf{X}_C = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{X} = \mathbf{H}_n\mathbf{X}$. Idempotency follows from $\mathbf{1}_n^\top\mathbf{1}_n = n$:

$$\left(\mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n\right)^2 = \mathbf{I}_n - \tfrac{2}{n}\mathbf{J}_n + \tfrac{1}{n^2}\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)\mathbf{1}_n^\top = \mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n.$$

Then $\mathbf{X}_C^\top\mathbf{X}_C = \mathbf{X}^\top\mathbf{H}_n^\top\mathbf{H}_n\mathbf{X} = \mathbf{X}^\top\mathbf{H}_n\mathbf{X}$, and $\mathbf{X}_C^\top\mathbf{X}_C = \sum_i(\mathbf{x}_i-\bar{\mathbf{x}})(\mathbf{x}_i-\bar{\mathbf{x}})^\top$. A symmetric idempotent matrix has eigenvalues 0 or 1, so $\operatorname{rank}(\mathbf{H}_n) = \operatorname{tr}(\mathbf{H}_n) = n-1$. ■

3 Sample Geometry in p-Space

3.1 The data ellipse

Each $\mathbf{x}_i$ is a point in $\mathbb{R}^p$. The mean vector is the center of gravity of the points and the covariance matrix describes their spread. The **data ellipse**

$$\{\mathbf{x} \in \mathbb{R}^p : (\mathbf{x}-\bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{x}-\bar{\mathbf{x}}) = c\}$$

summarizes location, shape and size for fixed $c > 0$ (an ellipsoid for $p \geq 3$). The penguin plot uses $c = 5.991 = \chi^2_2(0.05)$. If the cloud is not elliptical, the mean and covariance alone describe it poorly.

Example 1.5: data ellipses

For $\mathbf{S}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$ the axes of the ellipse are the coordinate axes with half-length ratio $3:1$. $\mathbf{S}_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ has eigenvalues $9, 1$ with eigenvectors $(1,1)^\top/\sqrt{2}$, $(1,-1)^\top/\sqrt{2}$: the same ellipse rotated by $45°$. Half-axis lengths are $\sqrt{c\lambda_j}$, so the ellipse reflects the size of variances and covariances.

3.2 Subspaces, orthogonal projection and projection scores

Definition 1.6: subspace and affine subspace

- **Subspace**: $\mathcal{L}_q = \operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\} = \{\mathbf{V}\mathbf{c} : \mathbf{c} \in \mathbb{R}^q\}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$ (linearly independent). It contains the origin.
- **Affine subspace**: $\mathcal{L}_q(\mathbf{x}) = \mathbf{x} + \mathcal{L}_q$, e.g. a line $\{\mathbf{x} + t\mathbf{v}\}$ or a plane $\{\mathbf{x} + c_1\mathbf{v}_1 + c_2\mathbf{v}_2\}$. It is not a vector space unless it contains the origin, but has the same shape as $\mathcal{L}_q$. Its representation (base point, spanning vectors) is not unique.

Definition 1.7: orthogonal projection and projection score

Projection onto $\operatorname{span}\{\mathbf{v}\}$ and the projection score (signed distance):

$$P_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}\mathbf{x} = s_{\mathbf{v}}(\mathbf{x})\frac{\mathbf{v}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}, \qquad s_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}^\top\mathbf{x}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}.$$

The projection matrix onto $\mathcal{L}_q$ is $\mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; for an orthonormal basis ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}_q$)

$$P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top\mathbf{x} = s_{\mathbf{v}_1}(\mathbf{x})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x})\mathbf{v}_q,$$

so $\mathbf{V}^\top\mathbf{x} = (s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))^\top$ are the coordinates in the subspace. The projection does not depend on the basis; the scores do.

Derivation: why $\mathbf{P}_{\mathbf{V}}$ is the orthogonal projection

$\mathbf{P}_{\mathbf{V}}^\top = \mathbf{P}_{\mathbf{V}}$ and $\mathbf{P}_{\mathbf{V}}^2 = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top = \mathbf{P}_{\mathbf{V}}$. Also $\mathbf{P}_{\mathbf{V}}\mathbf{x} \in \mathcal{L}_q$, and the residual satisfies $\mathbf{V}^\top(\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}) = \mathbf{0}$, so it is orthogonal to $\mathcal{L}_q$. By Pythagoras, for any $\mathbf{V}\mathbf{c}$, $\|\mathbf{x} - \mathbf{V}\mathbf{c}\|^2 = \|\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}\|^2 + \|\mathbf{P}_{\mathbf{V}}\mathbf{x} - \mathbf{V}\mathbf{c}\|^2 \geq \|\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}\|^2$: $\mathbf{P}_{\mathbf{V}}\mathbf{x}$ is the closest point of $\mathcal{L}_q$ to $\mathbf{x}$.

3.3 Projection onto an affine subspace and decomposition of observations

Projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is done in two steps:

(1) translate $\mathbf{x} \mapsto \mathbf{x} - \bar{\mathbf{x}}$ and project onto $\mathcal{L}_q$;

(2) translate back by $\bar{\mathbf{x}}$:

$$P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_{\mathbf{V}}(\mathbf{x} - \bar{\mathbf{x}}) = \bar{\mathbf{x}} + s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q.$$

Hence an observation splits into projection plus remainder, and by Pythagoras

$$\mathbf{x} = \bar{\mathbf{x}} + \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x}), \qquad d(\mathbf{x},\bar{\mathbf{x}})^2 = \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_q}(\mathbf{x})\|^2.$$

This decomposition is the starting point of dimension reduction (PCA).

Remark 1.8: geometric meaning of variance

Projection scores are tied to conditional variances. If $X_1 \sim N(0,\sigma_1^2)$ and $X_2 = X_1 + \epsilon_2$ with independent $\epsilon_2 \sim N(0,\sigma_2^2)$, then $\mathrm{Var}(X_2 \mid X_1) = \sigma_2^2$ and

$$\mathrm{Var}(X_1 \mid X_2) = \sigma_1^2 - \frac{\sigma_1^4}{\sigma_1^2+\sigma_2^2} = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2+\sigma_2^2}$$

(conditional distribution formula, Ch. 3): what remains after removing the projected part is the conditional variance.

3.4 Linear and affine transformations

A $q \times p$ matrix $\mathbf{L}$ defines the linear transformation $\mathbf{x} \mapsto \mathbf{L}\mathbf{x}$; $q > p$ only adds dimensions without information and is rarely used. Examples: the projection $\mathbf{x} \mapsto \mathbf{P}_{\mathbf{V}}\mathbf{x}$ and the score map $\mathbf{x} \mapsto \mathbf{S}_{\mathbf{V}}\mathbf{x}$, $\mathbf{S}_{\mathbf{V}} = (\mathbf{v}_1/\|\mathbf{v}_1\|, \ldots, \mathbf{v}_q/\|\mathbf{v}_q\|)^\top$. Projection onto an affine subspace is not linear (because of the translation) but an **affine transformation** $\mathbf{x} \mapsto \boldsymbol{m} + \mathbf{L}\mathbf{x}$.

Theorem 1.9: affine maps and standardization

1. Orthogonal projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is affine with $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{P}_{\mathbf{V}}\bar{\mathbf{x}}$, $\mathbf{L} = \mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; so is the score map $\mathbf{x} \mapsto (s_{\mathbf{v}_1}(\mathbf{x}-\bar{\mathbf{x}}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}-\bar{\mathbf{x}}))^\top$.
2. (Standardization) The data $\mathbf{x}_i' = \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})$ have sample mean $\mathbf{0}$ and unit sample variances, and $\mathbf{x}' = \mathbf{D}^{-1/2}\mathbf{x} - \mathbf{D}^{-1/2}\bar{\mathbf{x}}$ is affine.

Proof. (1) $P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_{\mathbf{V}}(\mathbf{x}-\bar{\mathbf{x}}) = (\bar{\mathbf{x}} - \mathbf{P}_{\mathbf{V}}\bar{\mathbf{x}}) + \mathbf{P}_{\mathbf{V}}\mathbf{x}$; the score vector is $\mathbf{S}_{\mathbf{V}}\mathbf{x} - \mathbf{S}_{\mathbf{V}}\bar{\mathbf{x}}$.

(2) $\bar{\mathbf{x}}' = \frac{1}{n}\sum_i \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathbf{0}$; the jth sample variance is $s_{jj}^{-1/2} s_{jj} s_{jj}^{-1/2} = 1$. ■

3.5 Rotation of coordinates

For $p = 2$ and $\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, put $\mathbf{v}_1 = \mathbf{R}_\theta\mathbf{e}_1$, $\mathbf{v}_2 = \mathbf{R}_\theta\mathbf{e}_2$; then $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ satisfies $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}_2$. Since $\mathbf{x} = \mathbf{V}\mathbf{V}^\top\mathbf{x} = [\mathbf{v}_1, \mathbf{v}_2](\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})^\top$, the coordinates of $\mathbf{x}$ in the new system are the projection scores $(\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})$. A linear map by an orthonormal basis is a rotation of coordinates.

4 Standardization, Sphering and Mahalanobis Distance

4.1 Univariate standardization and its limitation

For $N(\mu,\sigma^2)$ the distance of x from the center is better measured by the **standardized distance** $|x-\mu|/\sigma$ than by $|x-\mu|$. In the multivariate case, standardizing each variable, $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x}-\bar{\mathbf{x}})$,

removes scale differences but **ignores correlation**, so it does not give a distance that reflects the distribution.

Example 1.10: distances in bivariate data

$\bar{\mathbf{x}} = \mathbf{0}$, $\mathbf{S} = \begin{pmatrix} 640000 & 9600 \\ 9600 & 225 \end{pmatrix} = \begin{pmatrix} 800^2 & 800 \cdot 15 \cdot 0.8 \\ 800 \cdot 15 \cdot 0.8 & 15^2 \end{pmatrix}$ (correlation 0.8). For $\mathbf{y} = (-500, 10)^\top$, $\mathbf{z} = (1500, 15)^\top$: $d(\bar{\mathbf{x}}, \mathbf{y}) = 500.1$, $d(\bar{\mathbf{x}}, \mathbf{z}) = 1500.075$, so $\mathbf{z}$ looks farther. After standardization $\mathbf{y}' = (-0.625, 0.667)^\top$, $\mathbf{z}' = (1.875, 1.000)^\top$ and still $d(\mathbf{0}, \mathbf{y}') = 0.91 < 2.125 = d(\mathbf{0}, \mathbf{z}')$. Yet the scatterplot has many points near $\mathbf{z}$ and almost none near $\mathbf{y}$ ($\mathbf{y}$ goes against the positive correlation). After standardization the SD in the direction of $\mathbf{y}'$ is 0.36 and in the direction of $\mathbf{z}'$ is 1.31, not 1: standardization does not standardize spread in every direction.

4.2 Sphering

If the covariance is $\mathbf{I}_p$, the data ellipse is a sphere ($c = \|\mathbf{x} - \bar{\mathbf{x}}\|^2$) and spread is 1 in **every** direction.

Theorem 1.11: isotropy of sphered data

If $\bar{\mathbf{x}} = \mathbf{0}$ and $\mathbf{S} = \mathbf{I}_p$, then for any unit vector $\mathbf{v}$ the scores $S_{\mathbf{v}}(\mathbf{x}_i) = \mathbf{v}^\top \mathbf{x}_i$ have sample variance $\mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{v}^\top \mathbf{v} = 1$.

Definition 1.12: sphering transformation

A transformation after which the covariance is $\mathbf{I}_p$. For general $\mathbf{S}$ it is built in three steps.

1. **Rotation**: choose orthogonal $\mathbf{U}_S = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ so that $\mathbf{x}_i' = \mathbf{U}_S^\top \mathbf{x}_i$ has diagonal sample covariance $\mathbf{D}_S = \mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \mathrm{diag}(d_{11}, \ldots, d_{pp})$ (the eigendecomposition of $\mathbf{S}$).
2. **Standardization**: $\mathbf{x}' \mapsto \mathbf{D}_S^{-1/2} \mathbf{x}'$.
3. **Back-rotation**: $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{x}$.

Altogether $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \mathbf{x} = \mathbf{S}^{-1/2} \mathbf{x}$.

The sphered covariance is $\mathbf{I}_p$.

$$\begin{aligned} \mathbf{S}' &= \tfrac{1}{n-1} \textstyle\sum_i \mathbf{S}^{-1/2} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1/2} = \mathbf{S}^{-1/2} \mathbf{S} \mathbf{S}^{-1/2} \\ &= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{U}_S^\top = \mathbf{I}_p. \end{aligned}$$

■

Example 1.13: sphering in practice

(a) $\bar{\mathbf{x}} = (10, 10)^\top$, $\mathbf{S} = \mathrm{diag}(4, 9)$: the sphering matrix is $\mathbf{D}^{-1/2} = \mathrm{diag}(1/2, 1/3)$ and $\mathbf{S}' = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2} = \mathbf{I}_2$.

(b) $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. With $\mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (rotation by $\theta = \pi/4$),

$$\mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{D}_S,$$

so sphering is $\mathbf{x} \mapsto \mathbf{U}_S \mathrm{diag}(1/3, 1) \mathbf{U}_S^\top \mathbf{x}$. Applied to the data of Example 1.10, $\|\mathbf{y}' - \bar{\mathbf{x}}'\| = 2.65 > 2.18 = \|\mathbf{z}' - \bar{\mathbf{x}}'\|$: now $\mathbf{y}$ is correctly farther.

4.3 Mahalanobis distance

Definition 1.14: Mahalanobis (statistical) distance

For a distribution with mean $\bar{\mathbf{x}}$ and covariance $\mathbf{S}$, the Mahalanobis distance of $\mathbf{y}$ from $\bar{\mathbf{x}}$ is the Euclidean distance after sphering:

$$d_S(\mathbf{y}, \bar{\mathbf{x}}) = \|\mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})\|, \qquad d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{y} - \bar{\mathbf{x}}).$$

Proof. $d_S^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{U}_S \mathbf{D}_S^{-1/2} \underbrace{\mathbf{U}_S^\top \mathbf{U}_S}_{\mathbf{I}} \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})$ and $\mathbf{U}_S \mathbf{D}_S^{-1} \mathbf{U}_S^\top = \mathbf{S}^{-1}$. ■

Sphering and Mahalanobis distance are the multivariate counterparts of studentisation and standardized distance.

5 Eigendecomposition and Positive Definiteness

Motivation. A covariance matrix is symmetric and positive semidefinite, so it has an orthonormal eigenbasis. The eigenvectors are the natural axes of the data cloud and the eigenvalues the variances along them; PCA, Mahalanobis distance and the normal density all rest on this decomposition.

5.1 Eigendecomposition

For symmetric $\mathbf{A}$, $\mathbf{A}\mathbf{u} = \lambda \mathbf{u}$ ($\mathbf{u} \neq \mathbf{0}$) makes $\mathbf{u}$ an eigenvector and λ an eigenvalue: $\mathbf{A}$ only stretches or shrinks $\mathbf{u}$, by the factor λ.

Theorem 1.15: eigendecomposition of a symmetric matrix

A symmetric $\mathbf{A} \in \mathbb{R}^{p \times p}$ satisfies $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \sum_{i=1}^p \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ real.

- If $\mathbf{u}_i$ is an eigenvector so is $c\mathbf{u}_i$; normalized to length 1 it is unique up to sign (for distinct eigenvalues).
- Eigenvectors of distinct eigenvalues are orthogonal.
- With repeated eigenvalues the decomposition exists but $\mathbf{U}$ is not unique, e.g. $\sigma^2 \mathbf{I}_p = \mathbf{U}(\sigma^2 \mathbf{I})\mathbf{U}^\top$ for every orthogonal $\mathbf{U}$.

Orthogonality. From $\mathbf{A}\mathbf{u}_j = \lambda_j \mathbf{u}_j$, $\mathbf{u}_i^\top \mathbf{A}\mathbf{u}_j = \lambda_j \mathbf{u}_i^\top \mathbf{u}_j$; from $\mathbf{A}^\top = \mathbf{A}$, $\mathbf{u}_i^\top \mathbf{A} = \lambda_i \mathbf{u}_i^\top$, so $\mathbf{u}_i^\top \mathbf{A}\mathbf{u}_j = \lambda_i \mathbf{u}_i^\top \mathbf{u}_j$. Subtracting, $(\lambda_i - \lambda_j)\mathbf{u}_i^\top \mathbf{u}_j = 0$. ■

Example 1.16: eigenstructure of the sphering matrix

$\mathbf{A} = \mathbf{S}^{-1/2} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}$ (Example 1.13(b)), $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$: $\mathbf{A}\mathbf{u}_1 = \frac{1}{\sqrt{2}}(2/3 - 1/3, -1/3 + 2/3)^\top = \frac{1}{3}\mathbf{u}_1$; similarly $\mathbf{A}\mathbf{u}_2 = \mathbf{u}_2$. Sphering rescales each eigen-direction by $1/\sqrt{\lambda_j}$.

5.2 Inverse, square root, definiteness

If no eigenvalue is 0, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^\top$.

Definition 1.17: NND and PD

A symmetric $\mathbf{A}$ is **non-negative definite (NND)** if $\mathbf{x}^\top\mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x}$, and **positive definite (PD)** if $\mathbf{x}^\top\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Theorem 1.18: eigenvalues and definiteness

A symmetric $\mathbf{A}$ is PD iff all eigenvalues are positive, and NND iff all are non-negative.

Proof. With $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $\mathbf{y} = \mathbf{U}^\top\mathbf{x}$ (so $\mathbf{y} = \mathbf{0} \iff \mathbf{x} = \mathbf{0}$), $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \mathbf{y}^\top\mathbf{\Lambda}\mathbf{y} = \sum_i \lambda_i y_i^2$. If all $\lambda_i > 0$ and $\mathbf{x} \neq \mathbf{0}$ some $y_i \neq 0$ and the sum is positive; NND is the same. Conversely, if $\mathbf{A}$ is PD, $\lambda_i = \mathbf{u}_i^\top\mathbf{A}\mathbf{u}_i > 0$. ■

Theorem 1.19: the sample covariance is always NND, not necessarily PD

Proof. $\mathbf{z}^\top\mathbf{S}\mathbf{z} = \frac{1}{n-1}\sum_i\{(\mathbf{x}_i - \bar{\mathbf{x}})^\top\mathbf{z}\}^2 \geq 0$. But if the data lie in a lower-dimensional subspace of $\mathbb{R}^p$ (e.g. on a line), any $\mathbf{z} \neq \mathbf{0}$ orthogonal to it gives $\mathbf{z}^\top\mathbf{S}\mathbf{z} = 0$. ■

When n is much larger than p, $\mathbf{S}$ is PD in most cases and $\mathbf{S}^{-1}$ exists.

Definition 1.20: symmetric square root

For NND $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\mathbf{S}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$; for PD $\mathbf{S}$, $\mathbf{S}^{-1/2} = (\mathbf{S}^{-1})^{1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top$. $\mathbf{S}^{1/2}$ is symmetric with $\mathbf{S}^{1/2}\mathbf{S}^{1/2} = \mathbf{S}$.

5.3 Covariance matrices and ellipses

With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ the ellipse $E_S = \{\mathbf{x} : \mathbf{x}^\top\mathbf{S}^{-1}\mathbf{x} = 1\}$ reads

$$\frac{(\mathbf{u}_1^\top\mathbf{x})^2}{\lambda_1} + \cdots + \frac{(\mathbf{u}_p^\top\mathbf{x})^2}{\lambda_p} = 1,$$

so its principal axes are along $\mathbf{u}_i$ with half-lengths $\sqrt{\lambda_i}$. The eigendecomposition of a covariance matrix gives (1) the rotation $\mathbf{x} \mapsto \mathbf{U}^\top\mathbf{x}$ to uncorrelated coordinates and (2) the sphering $\mathbf{x} \mapsto \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top\mathbf{x}$.

Theorem 1.21: covariance of rotated data

If $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the rotated data $\mathbf{x}_i' = \mathbf{U}^\top\mathbf{x}_i$ have sample covariance $\mathbf{\Lambda}$.

Proof. $\frac{1}{n-1}\sum_i \mathbf{U}^\top(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\mathbf{U} = \mathbf{U}^\top\mathbf{S}\mathbf{U} = \mathbf{\Lambda}$. ■

The eigenvalues of a covariance matrix are the variances of mutually uncorrelated rotated coordinates — this is PCA.

6 Geometry of Principal Component Analysis

With many variables, inspecting all pairwise scatterplots is inefficient. PCA finds, in order, linear combinations of the original variables that explain the most variation, and summarizes the data by a few of them (details in Ch. 7).

6.1 Which direction to project on?

The scores $t_i(\mathbf{a}) = \mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$ along a unit vector $\mathbf{a}$ have sample variance $\frac{1}{n-1}\sum_i t_i(\mathbf{a})^2 = \mathbf{a}^\top\mathbf{S}\mathbf{a}$.

Definition 1.22: principal component directions

First direction $\mathbf{u}_1 \in \arg\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top\mathbf{S}\mathbf{a}$; later directions maximize variance subject to orthogonality to the earlier ones. The solutions are eigenvectors of $\mathbf{S}$.

$\mathbf{u}_1$ is the top eigenvector. With $\mathbf{z} = \mathbf{U}^\top\mathbf{a}$ ($\|\mathbf{z}\| = 1$), $\mathbf{a}^\top\mathbf{S}\mathbf{a} = \sum_j \lambda_j z_j^2 \le \lambda_1$, with equality at $\mathbf{a} = \mathbf{u}_1$. Under $z_1 = 0$ the maximum is λ_2 at $\mathbf{u}_2$, and so on. ■

6.2 Principal component scores

With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\lambda_1 \ge \cdots \ge \lambda_p \ge 0$, the score matrix is $\mathbf{T} = \mathbf{X}_C\mathbf{U} = (t_{ij})$, $t_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^\top\mathbf{u}_j$. Each column of $\mathbf{T}$ has mean 0 and $\frac{1}{n-1}\mathbf{T}^\top\mathbf{T} = \mathbf{U}^\top\mathbf{S}\mathbf{U} = \mathbf{\Lambda}$: the jth PC has sample variance λ_j and different PC scores are uncorrelated.

Example 1.23: scores in two subjects

Student	Math x_1	English x_2	PC1 score	PC2 score
A	55	65	−14.14	−7.07
B	65	55	−14.14	7.07
C	70	70	0	0
D	75	85	14.14	−7.07
E	85	75	14.14	7.07

$\bar{\mathbf{x}} = (70, 70)^\top$, $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$, $\lambda_1 = 200$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\lambda_2 = 50$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$.

$t_{i1} = \{(x_{i1} - 70) + (x_{i2} - 70)\}/\sqrt{2}$ is the overall level; $t_{i2} = \{(x_{i1} - 70) - (x_{i2} - 70)\}/\sqrt{2}$ is the relative difference between the subjects. A and B share the same PC1 score, so PC1 alone cannot distinguish their relative strengths.

6.3 Dimension reduction and reconstruction error

For student E, $\mathbf{x}_E = (85, 75)^\top$ and $t_{E1} = 10\sqrt{2}$. Reconstructing from PC1 gives $\hat{\mathbf{x}}_E = (70, 70)^\top + 10\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1)^\top = (80, 80)^\top$: the average level is kept, the difference between subjects is lost.

Theorem 1.24: reconstruction error

For the reconstruction $\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \sum_{j \le q} t_{ij}\mathbf{u}_j$ from the first q PCs,

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = (n-1) \sum_{j=q+1}^{p} \lambda_j.$$

Among all q-dimensional planes through $\bar{\mathbf{x}}$, PCA minimizes the reconstruction error, which is equivalent to maximizing the total projected variance.

Proof. $\mathbf{x}_i - \hat{\mathbf{x}}_i = \sum_{j>q} t_{ij}\mathbf{u}_j$, so $\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = \sum_{j>q} t_{ij}^2$ and $\sum_i t_{ij}^2 = (n-1)\lambda_j$. For any orthonormal $\mathbf{V} \in \mathbb{R}^{p\times q}$, Pythagoras gives $\sum_i \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{V}\mathbf{V}^\top(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S} - \mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})\}$, so minimizing the error is maximizing $\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})$, whose maximum $\lambda_1 + \cdots + \lambda_q$ follows from von Neumann's inequality (Ch. 2) or Ch. 7. ■

6.4 Proportion of variance, scale and caveats

- Since $\mathrm{tr}\mathbf{S} = \sum_k \lambda_k$, the proportion of variance explained by PC j is $\lambda_j / \sum_k \lambda_k$ and the cumulative proportion is $\sum_{j\le q} \lambda_j / \sum_k \lambda_k$ ($80\% = 200/250$ in the example). Choose q from the cumulative proportion or a scree plot.
- Covariance-based PCA depends on units (rescaling English to 1000 points multiplies its variance by 100). PCA of standardized data $z_{ik} = (x_{ik} - \bar{x}_k)/s_k$ is **correlation PCA**. If differences in variance are meaningful, standardizing is inappropriate.
- High explained variance does not mean good group separation. With $Y \in \{-1, 1\}$ equally likely, $X_1 = 10Z$, $X_2 = Y$, $Z \sim N(0,1) \perp\!\!\!\perp Y$: $\mathbf{\Sigma} = \mathrm{diag}(100, 1)$ and PC1 explains $100/101 \approx 99\%$, but $X_1 \perp\!\!\!\perp Y$ while X_2 alone separates the groups perfectly. PCA keeps variation, not predictive information.
- Still, for the Golub data ($p = 7129$), the PC1–PC2 plot, built without the labels, separates ALL/AML better than the original variables: PCA can be used to *explore* group differences.

7 Random Vectors and the Geometry of Variable Vectors

Motivation. Viewing each variable as a vector in $\mathbb{R}^n$ gives covariance and correlation a geometric meaning: lengths are standard deviations and angles are correlations. This picture makes projection arguments (regression, partial correlation) transparent.

7.1 Random vectors

The data $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top$ are a random sample from a population modeled by the joint distribution of $\mathbf{x} = (X_1, \ldots, X_p)^\top \sim f$.

Example 1.25: distributions of random vectors

1. Independent $X_i \sim \mathrm{Unif}(a_i, b_i)$: $f(\mathbf{x}) = \prod_i \frac{1}{b_i - a_i} \mathbf{1}\{x_i \in (a_i, b_i)\}$, uniform on a hypercube.
2. Independent $X_i \sim N(0,1)$: $f(\mathbf{x}) = (2\pi)^{-p/2} e^{-\|\mathbf{x}\|^2/2}$, the standard multivariate normal.
3. (X_1, X_2) uniform on $0 < x_1 < x_2 < 1$: $f = 2 \cdot \mathbf{1}\{0 < x_1 < x_2 < 1\}$. $X_1 \mid X_2 = x_2 \sim \mathrm{Unif}(0, x_2)$ depends on x_2, so they are not independent.

4. Bivariate normal with correlation ρ: $f = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\}$. Both marginals are $N(0,1)$ but the joint density is not their product: not independent.

7.2 Mean, variance and correlation through variable vectors

Viewing the variable vectors $\mathbf{x}_{(j)} \in \mathbb{R}^n$ (p vectors in n-space) gives a geometric reading of the statistics.

- **Mean**: $\bar{x}_j = \frac{1}{n}\mathbf{1}_n^\top \mathbf{x}_{(j)}$. The projection of $\mathbf{x}_{(j)}$ onto $\mathbf{1}_n$ is $P_{\mathbf{1}_n}(\mathbf{x}_{(j)}) = \mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)^{-1}\mathbf{1}_n^\top\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n$ with score $\sqrt{n}\,\bar{x}_j$.
- **Deviation vector**: $\mathbf{d}_j = \mathbf{x}_{(j)} - \bar{x}_j\mathbf{1}_n$; $\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n + \mathbf{d}_j$ is an orthogonal decomposition ($\mathbf{1}_n^\top\mathbf{d}_j = 0$).
- $s_j = \|\mathbf{d}_j\|/\sqrt{n-1}$, $s_j^2 = \|\mathbf{d}_j\|^2/(n-1)$, $s_{jk} = \mathbf{d}_j^\top\mathbf{d}_k/(n-1)$.
- **The sample correlation is the cosine of the angle between deviation vectors**: $r_{jk} = \dfrac{\mathbf{d}_j^\top\mathbf{d}_k}{\|\mathbf{d}_j\|\|\mathbf{d}_k\|} = \cos\theta_{jk}$.

A longer deviation vector means more variability. $\mathbf{d}_j = c\mathbf{d}_k$ with $c > 0$ ($\theta = 0$) gives $r = 1$, with $c < 0$ ($\theta = \pi$) gives $r = -1$; orthogonal deviation vectors give zero *linear* correlation.

Theorem 1.26: unbiasedness of $\bar{\mathbf{X}}$ and $\mathbf{S}$

If $\mathbf{X}_1, \ldots, \mathbf{X}_n$ is a random sample with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$,

$$\mathrm{E}(\bar{\mathbf{X}}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\bar{\mathbf{X}}) = \tfrac{1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}_n) = \tfrac{n-1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}.$$

Proof. $\mathrm{E}\bar{\mathbf{X}} = \frac{1}{n}\sum \mathrm{E}\mathbf{X}_j = \boldsymbol{\mu}$. $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}\sum_j\sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\}$, and terms with $j \neq \ell$ vanish by independence, giving $\frac{1}{n^2}\cdot n\boldsymbol{\Sigma}$. Since $\sum_j(\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top = \sum_j(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_j - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top$, its expectation is $n\boldsymbol{\Sigma} - \boldsymbol{\Sigma} = (n-1)\boldsymbol{\Sigma}$. ∎

7.3 Linear combinations of variables

For the new variable $z = a + \mathbf{b}^\top\mathbf{x} = a + \sum_j b_j x_j$ the variable vector is $\mathbf{z} = a\mathbf{1}_n + \sum_j b_j\mathbf{x}_{(j)}$, with

$$P_{\mathbf{1}_n}(\mathbf{z}) = \Big(a + \sum_j b_j\bar{x}_j\Big)\mathbf{1}_n, \qquad \mathbf{d}_z = \sum_j b_j\mathbf{d}_j = \mathbf{X}_C\mathbf{b}.$$

Theorem 1.27: sample variance and covariance of linear combinations

$$\widehat{\mathrm{Var}}(z) = \tfrac{1}{n-1}\mathbf{d}_z^\top\mathbf{d}_z = \mathbf{b}^\top\mathbf{S}\mathbf{b}, \qquad y = \mathbf{a}^\top\mathbf{x},\ z = \mathbf{b}^\top\mathbf{x} \ \Rightarrow\ \widehat{\mathrm{Cov}}(y,z) = \tfrac{1}{n-1}\mathbf{d}_y^\top\mathbf{d}_z = \mathbf{a}^\top\mathbf{S}\mathbf{b}.$$

For two variables, $y = b\,x_1 + c\,x_2$ has $\widehat{\mathrm{Var}}(y) = \{b^2\|\mathbf{d}_1\|^2 + c^2\|\mathbf{d}_2\|^2 + 2bc\|\mathbf{d}_1\|\|\mathbf{d}_2\|\cos\theta\}/(n-1)$.

8 Dimension and Rank of the Data

Motivation. If $n \leq p$, the sample covariance matrix is singular and cannot be inverted. The rank result here explains when classical multivariate procedures break down.

- Linear independence: $\sum c_i \mathbf{x}_i = \mathbf{0} \Rightarrow$ all $c_i = 0$. The rank is the maximal number of linearly independent columns (= rows). $\operatorname{rank}(\mathbf{X}) \leq \min(n, p)$.
- $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\top) = \operatorname{rank}(\mathbf{A}^\top \mathbf{A})$, hence $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C^\top \mathbf{X}_C) = \operatorname{rank}(\mathbf{X}_C)$.
- All deviation vectors are orthogonal to $\mathbf{1}_n$, so they lie in an $(n-1)$-dimensional subspace: $\operatorname{rank}(\mathbf{X}_C) \leq \min(n-1, p)$.
- If $p > n$, $\operatorname{rank}(\mathbf{S}) \leq n - 1 < p$: $\mathbf{S}$ is always rank-deficient.

Theorem 1.28: rank-deficiency property

For the covariance matrix $\mathbf{S}$ of $\mathbf{X} \in \mathbb{R}^{n \times p}$ the following are equivalent.

1. $\mathbf{S}$ is rank-deficient.
2. There is $\mathbf{a} \neq \mathbf{0}$ with $\mathbf{a}^\top (\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i.
3. All projection scores coincide: $s_{\mathbf{a}}(\mathbf{x}_i) = s_{\mathbf{a}}(\mathbf{x}_j)$ for all i, j.
4. $\operatorname{span}\{\mathbf{d}_1, \ldots, \mathbf{d}_p\}$ has dimension less than p.
5. Some deviation vector is a linear combination of the others.
6. At least one eigenvalue of $\mathbf{S}$ is 0.

Proof. (1)⇔(2): $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C^\top) = r < p$ iff the columns $\{\mathbf{x}_i - \bar{\mathbf{x}}\}$ of $\mathbf{X}_C^\top$ span a space of dimension $r < p$, iff a nonzero $\mathbf{a}$ in its orthogonal complement exists. (2)⇔(3): $\mathbf{a}^\top (\mathbf{x}_i - \bar{\mathbf{x}}) = 0 \iff \mathbf{a}^\top \mathbf{x}_i = \mathbf{a}^\top \bar{\mathbf{x}}$ for all i. (1)⇔(4): $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C) = \dim \operatorname{span}\{\mathbf{d}_j\}$. (4)⇔(5): dimension below p means $\{\mathbf{d}_j\}$ is linearly dependent. (2)⇒(6): $\mathbf{S}\mathbf{a} = \frac{1}{n-1} \mathbf{X}_C^\top \mathbf{X}_C \mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$. (6)⇒(2): $\mathbf{S}\mathbf{u} = \mathbf{0}$ gives $\mathbf{u}^\top \mathbf{S} \mathbf{u} = \frac{1}{n-1} \|\mathbf{X}_C \mathbf{u}\|^2 = 0$, so $\mathbf{X}_C \mathbf{u} = \mathbf{0}$. ■

If $\mathbf{S}$ is full rank, every direction carries independent variation. If it is rank-deficient, some direction carries no information at all (an exact linear relation among the variables), the sphering $\mathbf{x} \mapsto \mathbf{S}^{-1/2} \mathbf{x}$ is undefined, and statistical analysis becomes less reliable. In high dimensions ($p > n$) $\mathbf{S}$ is always rank-deficient: $\operatorname{rank}(\mathbf{S}) \leq \min\{n - 1, p\} < p$.

Chapter 2. Matrix, Calculus and Probability Tools

Matrix identities, eigenvalue inequalities, matrix calculus, optimization, convergence

This chapter collects the tools used in later proofs (MLE, Wishart, T^2, large-sample theory).

1 Basic Matrix Operations

Motivation. This chapter collects the linear-algebra and probability tools used throughout, so later proofs can cite them. We start with trace, determinant, inverses of partitioned matrices and Kronecker products.

Definition 2.1: basic facts

(i) $(\mathbf{AB})_{ik} = \sum_j a_{ij}b_{jk}$; $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$; a 1×1 matrix equals its transpose.

(ii) $\operatorname{tr}(\mathbf{C}) = \sum_i c_{ii}$ is linear.

(iii) $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$, so $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$.

(iv) (Leibniz) $|\mathbf{A}| = \sum_\sigma \operatorname{sgn}(\sigma)\prod_i a_{i\sigma(i)}$; hence $|\mathbf{A}|$ is a polynomial in the entries, continuous and infinitely differentiable, and $\{|\mathbf{A}| \neq 0\}$ is open. $|\mathbf{I}_p| = 1$, $|\operatorname{diag}(d_i)| = \prod d_i$, $|c\mathbf{A}| = c^p|\mathbf{A}|$.

(v) (Adjugate) If $|\mathbf{A}| \neq 0$, $\mathbf{A}^{-1} = \operatorname{adj}(\mathbf{A})/|\mathbf{A}|$, so $\mathbf{A} \mapsto \mathbf{A}^{-1}$ is continuous and differentiable on $\{|\mathbf{A}| \neq 0\}$.

(vi) (Cauchy–Schwarz) $|\sum_k a_k b_k| \le (\sum a_k^2)^{1/2}(\sum b_k^2)^{1/2}$.

Definition 2.2: Frobenius norm

$\|\mathbf{A}\|_F = (\sum_{i,j} a_{ij}^2)^{1/2}$ and $\|\mathbf{x}\| = \|\mathbf{x}\|_F$ for vectors. Example: $\mathbf{M} = \begin{pmatrix} -0.1 & 0.3 \\ 0.2 & -0.2 \end{pmatrix}$ has $\|\mathbf{M}\|_F = \sqrt{0.18} \approx 0.424$. Estimation errors of the sample moments can be measured by $E_{\bar u} = \|\bar{\mathbf{u}} - \boldsymbol{\mu}\|_2$, $E_S = \|\mathbf{S} - \boldsymbol{\Sigma}\|_F$, $E_R = \|\mathbf{R} - \mathbf{R}_0\|_F$.

Theorem 2.3: basic identities

For $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{y} \in \mathbb{R}^q$, $\mathbf{A} \in \mathbb{R}^{p\times q}$, $\mathbf{B} \in \mathbb{R}^{q\times p}$:

(i) $\mathbf{x}^\top\mathbf{A}\mathbf{y} = \sum_{i,j} a_{ij}x_i y_j$; if $\mathbf{A}$ is diagonal, $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i a_{ii}x_i^2$.

(ii) $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$.

(iii) $\|\mathbf{A}\|_F^2 = \operatorname{tr}(\mathbf{A}^\top\mathbf{A})$ and $\|\mathbf{x}\|^2 = \mathbf{x}^\top\mathbf{x}$.

Proof. (i) $(\mathbf{Ay})_i = \sum_j a_{ij}y_j$, so $\mathbf{x}^\top\mathbf{Ay} = \sum_i x_i(\mathbf{Ay})_i$.

(ii) $\operatorname{tr}(\mathbf{AB}) = \sum_i\sum_j a_{ij}b_{ji} = \sum_j\sum_i b_{ji}a_{ij} = \operatorname{tr}(\mathbf{BA})$.

(iii) $(\mathbf{A}^\top\mathbf{A})_{jj} = \sum_i a_{ij}^2$; sum over j. ∎

Theorem 2.4: first-order expansion of the determinant

$|\mathbf{I}_p + t\mathbf{N}| = 1 + t\,\mathrm{tr}(\mathbf{N}) + O(t^2)$ as $t \to 0$.

Proof. In the Leibniz formula a non-identity σ moves at least two indices, so its term is $O(t^2)$. The identity contributes $\prod_i (1 + t n_{ii}) = 1 + t\sum_i n_{ii} + O(t^2)$. ■

Definition 2.5: open sets and continuity

With distance $\|\mathbf{B} - \mathbf{A}\|_F$: U is open iff every $\mathbf{A} \in U$ has a ball of radius $r > 0$ inside U; f is continuous at $\mathbf{A}$ iff $\forall \epsilon\, \exists \delta$ with $\|\mathbf{B} - \mathbf{A}\|_F < \delta \Rightarrow \|f(\mathbf{B}) - f(\mathbf{A})\|_F < \epsilon$. If every entry of a map is continuous, so is the map ($\epsilon/\sqrt{m}$ argument).

1.1 Block-matrix formulas

Used for conditional distributions, the Wishart theorem and likelihood ratio tests.

Theorem 2.6: block determinant and inverse

For $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$ with $\mathbf{A}_{22}$ invertible,

$$|\mathbf{A}| = |\mathbf{A}_{22}|\,|\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}|\,|\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}| \quad (\mathbf{A}_{11} \text{ invertible}).$$

With the Schur complement $\mathbf{A}_{11\cdot 2} := \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$,

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11\cdot 2}^{-1} & -\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1} & \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{A}_{11\cdot 2}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{pmatrix};$$

in particular $(\mathbf{A}^{-1})_{11} = \mathbf{A}_{11\cdot 2}^{-1}$.

Proof. Block Gaussian elimination:

$$\begin{pmatrix} \mathbf{I} & -\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \mathbf{A} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11\cdot 2} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix}.$$

The outer factors have determinant 1, giving the determinant formula; inverting both sides gives $\mathbf{A}^{-1}$. ■

Theorem 2.7: matrix determinant lemma and Woodbury identity

$$|\mathbf{A} + \mathbf{u}\mathbf{v}^\top| = |\mathbf{A}|(1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}), \qquad |\mathbf{I}_p + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u},$$
$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{A}^{-1}.$$

Proof. Compute $\begin{vmatrix} \mathbf{A} & -\mathbf{u} \\ \mathbf{v}^\top & 1 \end{vmatrix}$ with both forms of the block formula: $|\mathbf{A}|(1+\mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}) = 1\cdot|\mathbf{A}+\mathbf{u}\mathbf{v}^\top|$. For Woodbury, multiply the right side by $\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V}$ and check that the product is $\mathbf{I}$. ■

2 Eigenvalues and Positive Definiteness

Motivation. Most multivariate statistics are functions of eigenvalues of covariance matrices; their basic properties are recorded here.

Theorem 2.8: spectral decomposition and consequences

Symmetric $\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}(\mathbf{A})\mathbf{U}^\top = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$, $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_p(\mathbf{A})$ ($\lambda_i(\mathbf{A})$ denotes the ith largest eigenvalue).

$$|\mathbf{A}| = \prod_i \lambda_i(\mathbf{A}), \qquad \operatorname{tr}(\mathbf{A}) = \sum_i \lambda_i(\mathbf{A}), \qquad \mathbf{A} \succ 0 \iff \text{all } \lambda_i(\mathbf{A}) > 0.$$

Proof. $|\mathbf{U}||\mathbf{U}^\top| = |\mathbf{U}\mathbf{U}^\top| = 1$ gives $|\mathbf{A}| = |\boldsymbol{\Lambda}(\mathbf{A})|$; $\operatorname{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top) = \operatorname{tr}(\boldsymbol{\Lambda}\mathbf{U}^\top\mathbf{U}) = \operatorname{tr}\boldsymbol{\Lambda}$. Definiteness follows from $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum_i \lambda_i(\mathbf{u}_i^\top\mathbf{x})^2$. ■

Theorem 2.9: smallest eigenvalue

For symmetric $\mathbf{A}, \mathbf{B}$:
(i) $\lambda_p(\mathbf{A}) = \min_{\|\mathbf{x}\|=1} \mathbf{x}^\top\mathbf{A}\mathbf{x}$;
(ii) $|\lambda_p(\mathbf{A}) - \lambda_p(\mathbf{B})| \leq \|\mathbf{A} - \mathbf{B}\|_F$;
(iii) $\{\mathbf{A} \succ 0\}$ is convex and open in the space of symmetric matrices.

Proof. (i) With $\mathbf{z} = \mathbf{U}^\top\mathbf{x}$, $\|\mathbf{z}\| = 1$ and $\mathbf{x}^\top\mathbf{A}\mathbf{x} = \sum \lambda_i z_i^2 \geq \lambda_p$, equality at $\mathbf{x} = \mathbf{u}_p$.

(ii) For $\|\mathbf{x}\| = 1$, Cauchy–Schwarz gives $|\mathbf{x}^\top\mathbf{A}\mathbf{x} - \mathbf{x}^\top\mathbf{B}\mathbf{x}| = |\sum_{ij}(a_{ij} - b_{ij})x_i x_j| \leq \|\mathbf{A} - \mathbf{B}\|_F(\sum_{ij} x_i^2 x_j^2)^{1/2} = \|\mathbf{A} - \mathbf{B}\|_F$; minimize both sides over $\mathbf{x}$.

(iii) Convex: $\mathbf{x}^\top\{(1-s)\mathbf{A}_0 + s\mathbf{A}_1\}\mathbf{x} > 0$. Open: if $\mathbf{A} \succ 0$ and $\|\mathbf{B} - \mathbf{A}\|_F < \lambda_p(\mathbf{A})$, (ii) gives $\lambda_p(\mathbf{B}) \geq \lambda_p(\mathbf{A}) - \|\mathbf{A} - \mathbf{B}\|_F > 0$. ■

Theorem 2.10: symmetric square root

For $\mathbf{A} \succ 0$, $\mathbf{A}^{\pm 1/2} = \mathbf{U}\boldsymbol{\Lambda}^{\pm 1/2}\mathbf{U}^\top$ are symmetric, $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{A}$, $\mathbf{A}^{-1/2}\mathbf{A}^{-1/2} = \mathbf{A}^{-1}$, $\mathbf{A}^{-1/2}\mathbf{A}\mathbf{A}^{-1/2} = \mathbf{I}_p$. $\mathbf{A}^{-1} = \mathbf{U}\boldsymbol{\Lambda}^{-1}\mathbf{U}^\top \succ 0$ and $\lambda_i(\mathbf{A}^{-1}) = 1/\lambda_{p+1-i}(\mathbf{A})$ with eigenvector $\mathbf{u}_{p+1-i}$.

3 Eigenvalue Inequalities

Motivation. PCA, canonical correlation and discriminant analysis all maximize a ratio of quadratic forms. The extremal characterization of eigenvalues (Rayleigh quotient, Courant–Fischer) is the common tool that solves them.

Definition 2.11: Birkhoff's theorem and the rearrangement inequality

(i) A non-negative matrix with all row and column sums 1 (doubly stochastic) is a convex combination of permutation matrices.
(ii) If $a_1 \geq \cdots \geq a_p$ and $b_1 \geq \cdots \geq b_p$, then for every permutation σ, $\sum_i a_i b_{p+1-i} \leq \sum_i a_i b_{\sigma(i)} \leq \sum_i a_i b_i$.

Theorem 2.12: von Neumann's trace inequality

For symmetric $\mathbf{A} = \sum_i \lambda_i(\mathbf{A})\mathbf{u}_i\mathbf{u}_i^\top$, $\mathbf{B} = \sum_j \lambda_j(\mathbf{B})\mathbf{v}_j\mathbf{v}_j^\top$,

$$\sum_{i=1}^{p} \lambda_{p+1-i}(\mathbf{A})\lambda_i(\mathbf{B}) \leq \operatorname{tr}(\mathbf{A}\mathbf{B}) \leq \sum_{i=1}^{p} \lambda_i(\mathbf{A})\lambda_i(\mathbf{B}).$$

The upper bound is attained when $\mathbf{v}_i = \mathbf{u}_i$ and the lower bound when $\mathbf{v}_i = \mathbf{u}_{p+1-i}$.

Proof. Step 1. $\operatorname{tr}(\mathbf{AB}) = \sum_{i,j} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) w_{ij}$, $w_{ij} = (\mathbf{u}_i^\top \mathbf{v}_j)^2 \geq 0$. Since $\{\mathbf{u}_i\}, \{\mathbf{v}_j\}$ are orthonormal bases, $\sum_j w_{ij} = \|\mathbf{u}_i\|^2 = 1$ and $\sum_i w_{ij} = \|\mathbf{v}_j\|^2 = 1$: $\mathbf{W}$ is doubly stochastic. **Step 2.** By Birkhoff $\mathbf{W}$ is a convex combination of permutation matrices; $\operatorname{tr}(\mathbf{AB})$ is linear in $\mathbf{W}$, so it lies between the minimum and maximum of $\sum_i \lambda_i(\mathbf{A})\lambda_{\sigma(i)}(\mathbf{B})$. **Step 3.** By rearrangement the maximum is at $\sigma(i) = i$, the minimum at $\sigma(i) = p+1-i$. **Step 4.** If $\mathbf{v}_i = \mathbf{u}_i$ (resp. $\mathbf{u}_{p+1-i}$), $\mathbf{W}$ is the identity (resp. reversal) permutation and the bound is attained. ■

Theorem 2.13: an elementary inequality

$x - \log x \geq 1$ for $x > 0$, with equality iff $x = 1$.

Proof. $\phi(x) = x - \log x$ has $\phi'(x) = 1 - 1/x$, negative on $(0,1)$, positive on $(1,\infty)$. For $x \neq 1$ the mean value theorem gives $\phi(x) - \phi(1) = \phi'(\xi)(x-1) > 0$ with ξ between 1 and x. ■

4 Vector Calculus

Motivation. Maximum likelihood for the multivariate normal requires derivatives with respect to vectors and matrices, including $\log\det \mathbf{\Sigma}$. These rules are collected here.

Definition 2.14: gradient, Jacobian, Hessian

For $f : \mathbb{R}^p \to \mathbb{R}$, $\partial f/\partial \mathbf{x} = (\partial f/\partial x_1, \ldots, \partial f/\partial x_p)^\top$. The Jacobian of $\mathbf{g} : \mathbb{R}^p \to \mathbb{R}^q$ is the $q \times p$ matrix $(\partial g_i/\partial x_j)$. The Hessian $\partial^2 f/\partial \mathbf{x}\partial \mathbf{x}^\top$ is the Jacobian of the gradient; it is symmetric when second partials are continuous (Schwarz's theorem).

Theorem 2.15: chain rule

For $\mathbf{y} = \mathbf{g}(\mathbf{x})$ with $q \times p$ Jacobian $\mathbf{J} = \partial \mathbf{g}/\partial \mathbf{x}^\top$,

$$\frac{\partial f(\mathbf{y})}{\partial \mathbf{x}} = \mathbf{J}^\top \frac{\partial f}{\partial \mathbf{y}}, \qquad \frac{\partial^2 f(\mathbf{y})}{\partial \mathbf{x}\partial \mathbf{x}^\top} = \mathbf{J}^\top \frac{\partial^2 f}{\partial \mathbf{y}\partial \mathbf{y}^\top}\mathbf{J} + \sum_{i=1}^{q} \frac{\partial f}{\partial y_i}\frac{\partial^2 g_i}{\partial \mathbf{x}\partial \mathbf{x}^\top}.$$

For an affine map $\mathbf{g}(\mathbf{x}) = \mathbf{B}\mathbf{x} + \mathbf{b}$, $\mathbf{J} = \mathbf{B}$ and the last sum vanishes.

Theorem 2.16: derivatives of quadratic forms and basic maps

For symmetric $\mathbf{A}$, $\dfrac{\partial\, \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}$ and $\dfrac{\partial^2\, \mathbf{x}^\top \mathbf{A}\mathbf{x}}{\partial \mathbf{x}\partial \mathbf{x}^\top} = 2\mathbf{A}$. Also $\partial \mathbf{x}/\partial \mathbf{x}^\top = \mathbf{I}_p$, $\partial(\mathbf{B}\mathbf{x} + \mathbf{b})/\partial \mathbf{x}^\top = \mathbf{B}$, and $\partial(\mathbf{x} - \boldsymbol{\mu})/\partial \boldsymbol{\mu}^\top = -\mathbf{I}_p$.

Proof. $\mathbf{x}^\top \mathbf{A}\mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$, so $\partial/\partial x_k = \sum_j a_{kj} x_j + \sum_i a_{ik} x_i = 2(\mathbf{A}\mathbf{x})_k$; differentiating in x_l gives $2a_{kl}$. ■

5 Matrix Calculus

To optimize over covariance matrices we differentiate with respect to a matrix: matrix gradients give first-order conditions and directional derivatives give second-order conditions along symmetric

directions.

Definition 2.17: directional derivatives and matrix gradient

For f on an open set C, $\boldsymbol{\Theta} \in C$ and a direction $\mathbf{H}$,

$$Df(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d}{dt} f(\boldsymbol{\Theta} + t\mathbf{H})\Big|_{t=0}, \qquad D^2 f(\boldsymbol{\Theta})[\mathbf{H}] = \frac{d^2}{dt^2} f(\boldsymbol{\Theta} + t\mathbf{H})\Big|_{t=0}.$$

The matrix gradient $\partial f / \partial \mathbf{X}$ has (i,j) entry $\partial f / \partial x_{ij}$.

Theorem 2.18: directional derivative and gradient

In $\mathbb{R}^p$: $Df(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \frac{\partial f}{\partial \boldsymbol{\mu}}$, $D^2 f(\boldsymbol{\mu})[\mathbf{h}] = \mathbf{h}^\top \frac{\partial^2 f}{\partial \boldsymbol{\mu} \partial \boldsymbol{\mu}^\top} \mathbf{h}$. For matrices: $Df(\mathbf{X})[\mathbf{H}] = \operatorname{tr}\Big\{\Big(\frac{\partial f}{\partial \mathbf{X}}\Big)^\top \mathbf{H}\Big\} = \sum_{i,j} \frac{\partial f}{\partial x_{ij}} h_{ij}$.

Theorem 2.19: derivatives of basic matrix functions

For $\boldsymbol{\Sigma} \succ 0$, symmetric $\mathbf{H}$ and $p \times p$ $\mathbf{B}$:

$$D \log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}), \qquad D^2 \log|\boldsymbol{\Sigma}|[\mathbf{H}] = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H}),$$

$$D \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = \operatorname{tr}(\mathbf{H}\mathbf{B}), \qquad D^2 \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})[\mathbf{H}] = 0,$$

$$\frac{\partial \log|\boldsymbol{\Sigma}|}{\partial \boldsymbol{\Sigma}} = \boldsymbol{\Sigma}^{-1}, \qquad \frac{\partial \operatorname{tr}(\boldsymbol{\Sigma}\mathbf{B})}{\partial \boldsymbol{\Sigma}} = \mathbf{B}^\top.$$

D and D^2 are linear.

Proof. Let $\mathbf{M} = \boldsymbol{\Sigma}^{-1/2}\mathbf{H}\boldsymbol{\Sigma}^{-1/2} = \mathbf{V}\boldsymbol{\Lambda}(\mathbf{M})\mathbf{V}^\top$ (symmetric). Then $\boldsymbol{\Sigma} + t\mathbf{H} = \boldsymbol{\Sigma}^{1/2}(\mathbf{I} + t\mathbf{M})\boldsymbol{\Sigma}^{1/2}$ and $\mathbf{I} + t\mathbf{M}$ has eigenvalues $1 + t\lambda_k(\mathbf{M})$, so for small $|t|$

$$\log|\boldsymbol{\Sigma} + t\mathbf{H}| = \log|\boldsymbol{\Sigma}| + \sum_k \log\{1 + t\lambda_k(\mathbf{M})\}.$$

The first and second derivatives at $t = 0$ are $\sum_k \lambda_k(\mathbf{M}) = \operatorname{tr}\mathbf{M} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H})$ and $-\sum_k \lambda_k(\mathbf{M})^2 = -\operatorname{tr}(\mathbf{M}^2) = -\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}\boldsymbol{\Sigma}^{-1}\mathbf{H})$. $\operatorname{tr}\{(\boldsymbol{\Sigma} + t\mathbf{H})\mathbf{B}\}$ is linear in t. Gradients: for any $\mathbf{H}$, $|\boldsymbol{\Sigma} + t\mathbf{H}| = |\boldsymbol{\Sigma}||\mathbf{I} + t\boldsymbol{\Sigma}^{-1}\mathbf{H}| = |\boldsymbol{\Sigma}|\{1 + t\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{H}) + O(t^2)\}$, so $D \log|\boldsymbol{\Sigma}|[\mathbf{H}] = \operatorname{tr}\{(\boldsymbol{\Sigma}^{-1})^\top \mathbf{H}\}$; compare with Theorem 2.18. Likewise $\operatorname{tr}(\mathbf{H}\mathbf{B}) = \operatorname{tr}\{(\mathbf{B}^\top)^\top \mathbf{H}\}$. ■

Theorem 2.20: symmetric matrix variables

If $\mathbf{X}$ is symmetric and $\mathbf{G} = \partial f / \partial \mathbf{X}$ is symmetric, then $Df(\mathbf{X})[\mathbf{H}] = 0$ for all symmetric $\mathbf{H}$ iff $\mathbf{G} = \mathbf{0}$.

Proof. $\Leftarrow$ is clear. $\Rightarrow$: take $\mathbf{H} = \mathbf{G}$: $0 = \operatorname{tr}(\mathbf{G}^\top \mathbf{G}) = \|\mathbf{G}\|_F^2$. ■

6 Optimization

Motivation. Several estimators are defined as maximizers subject to constraints. Lagrange multipliers and the trace inequality used for the normal MLE are stated here.

Theorem 2.21: convexity and unique minimizer

Let C be open and convex and suppose $D^2 f(\boldsymbol{\Theta})[\mathbf{H}] > 0$ for all $\boldsymbol{\Theta} \in C$, $\mathbf{H} \neq \mathbf{0}$. Then (i) f is strictly convex on C;
(ii) a point $\hat{\boldsymbol{\Theta}}$ with $Df(\hat{\boldsymbol{\Theta}})[\mathbf{H}] = 0$ for all $\mathbf{H}$ is the unique global minimizer;
(iii) for $C = \mathbb{R}^p$ these read: Hessian $\succ 0$ and gradient $= \mathbf{0}$.

Proof. $g(t) = f(\boldsymbol{\Theta}_0 + t(\boldsymbol{\Theta}_1 - \boldsymbol{\Theta}_0))$ has $g''(t) = D^2 f(\cdot)[\mathbf{H}] > 0$, so g is strictly convex.

(ii) With $\boldsymbol{\Theta}_0 = \hat{\boldsymbol{\Theta}}$, $g'(0) = 0$ and g' is strictly increasing, so $g'(t) > 0$ on $(0, 1]$ and $f(\boldsymbol{\Theta}_1) = g(1) > g(0) = f(\hat{\boldsymbol{\Theta}})$ by the mean value theorem. ■

Theorem 2.22: transformations that keep the optimizer; profiling

For $a > 0$ and a constant c: $\arg\min\{af + c\} = \arg\min f$, $\arg\max f = \arg\min(-f)$, $\arg\max L = \arg\max \log L$ for $L > 0$, and for a bijection ϕ, $\arg\min_\theta f(\theta) = \phi^{-1}(\arg\min_\eta f \circ \phi^{-1}(\eta))$. **Profiling**: if $\hat{\theta}(\eta)$ minimizes $\ell(\cdot, \eta)$ for each η and $\hat{\eta}$ minimizes the profile $p(\eta) = \ell(\hat{\theta}(\eta), \eta)$, then $(\hat{\theta}(\hat{\eta}), \hat{\eta})$ minimizes ℓ jointly: $\ell(\theta, \eta) \geq \ell(\hat{\theta}(\eta), \eta) = p(\eta) \geq p(\hat{\eta})$.

The MLE (Ch. 4) first profiles out $\boldsymbol{\mu}$ and then minimizes over $\boldsymbol{\Sigma}$.

7 Probability and Convergence

Motivation. Large-sample results need convergence in probability and distribution for random vectors; the multivariate CLT and Slutsky's lemma are the tools.

Definition 2.23: standard results from mathematical statistics

(S0) $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$: $P(\|\mathbf{Y}_n - \mathbf{c}\| > \epsilon) \to 0$ for all $\epsilon > 0$. $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$: $\mathrm{E}g(\mathbf{Y}_n) \to \mathrm{E}g(\mathbf{Y})$ for all bounded continuous g.

(S1) Markov: for $W \geq 0$, $P(W \geq a) \leq \mathrm{E}W/a$.

(S2) $\mathrm{E}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\mathrm{E}\mathbf{Y} + \mathbf{b}$, $\mathrm{Cov}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\mathrm{Cov}(\mathbf{Y})\mathbf{A}^\top$; for independent $\mathbf{Y}_j$, $\mathrm{Cov}(\sum \mathbf{Y}_j) = \sum \mathrm{Cov}(\mathbf{Y}_j)$; $\mathrm{E}\|\mathbf{Y} - \mathrm{E}\mathbf{Y}\|^2 = \mathrm{tr}\mathrm{Cov}(\mathbf{Y})$.

(S3) Weak law of large numbers; (S4) univariate CLT.

(S5) Continuous mapping theorem (CMT): $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, g continuous $\Rightarrow g(\mathbf{Y}_n) \xrightarrow{d} g(\mathbf{Y})$; $\mathbf{Y}_n \xrightarrow{p} \mathbf{c}$, g continuous at $\mathbf{c} \Rightarrow g(\mathbf{Y}_n) \xrightarrow{p} g(\mathbf{c})$.

(S6) Slutsky: $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y}$, $\mathbf{W}_n \xrightarrow{p} \mathbf{c} \Rightarrow (\mathbf{Y}_n, \mathbf{W}_n) \xrightarrow{d} (\mathbf{Y}, \mathbf{c})$.

(S7) Componentwise convergence in probability to constants implies joint convergence.

Theorem 2.24: vector Markov inequality and matrix convergence

$P(\|\mathbf{Y}\| > \epsilon) \leq \mathrm{E}\|\mathbf{Y}\|^2/\epsilon^2$. For random matrices, $\mathbf{A}_n \xrightarrow{p} \mathbf{A}$ (entrywise) $\iff \|\mathbf{A}_n - \mathbf{A}\|_F \xrightarrow{p} 0$.

Proof. $\{\|\mathbf{Y}\| > \epsilon\} \subset \{\|\mathbf{Y}\|^2 \geq \epsilon^2\}$; apply Markov. Matrices: $\|\mathbf{A}_n - \mathbf{A}\|_F^2 \leq pq \max_{ij}(\mathbf{A}_n - \mathbf{A})_{ij}^2$, so $\{\|\mathbf{A}_n - \mathbf{A}\|_F > \epsilon\} \subset \bigcup_{ij}\{|(\mathbf{A}_n - \mathbf{A})_{ij}| > \epsilon/\sqrt{pq}\}$ and a union bound gives convergence; conversely $|(\mathbf{A}_n - \mathbf{A})_{ij}| \leq \|\mathbf{A}_n - \mathbf{A}\|_F$. ■

Theorem 2.25: characteristic functions and the Cramér–Wold device

$\varphi_{\mathbf{Y}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top\mathbf{Y})$.
(i) Uniqueness: $\varphi_{\mathbf{Y}} \equiv \varphi_{\mathbf{W}} \iff \mathbf{Y} \overset{d}{=} \mathbf{W}$.
(ii) Lévy continuity: $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \iff \varphi_{\mathbf{Y}_n}(\mathbf{t}) \to \varphi_{\mathbf{Y}}(\mathbf{t})$ for all $\mathbf{t}$. **Cramér–Wold**:
(i) the distribution of $\mathbf{Y}$ is determined by those of all $\mathbf{t}^\top\mathbf{Y}$;
(ii) $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \iff \mathbf{t}^\top\mathbf{Y}_n \xrightarrow{d} \mathbf{t}^\top\mathbf{Y}$ for every $\mathbf{t}$.

Proof. The key identity is $\varphi_{\mathbf{t}^\top\mathbf{Y}}(1) = \mathrm{E}e^{i\mathbf{t}^\top\mathbf{Y}} = \varphi_{\mathbf{Y}}(\mathbf{t})$.

(i) If $\mathbf{t}^\top\mathbf{Y} \overset{d}{=} \mathbf{t}^\top\mathbf{W}$, then $\varphi_{\mathbf{Y}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top\mathbf{Y}}(1) = \varphi_{\mathbf{t}^\top\mathbf{W}}(1) = \varphi_{\mathbf{W}}(\mathbf{t})$.

(ii) $\Leftarrow$: Lévy for scalars gives $\varphi_{\mathbf{t}^\top\mathbf{Y}_n}(s) \to \varphi_{\mathbf{t}^\top\mathbf{Y}}(s)$; take $s = 1$. ■

"All one-dimensional linear projections determine the multivariate distribution" underlies the general definition of the multivariate normal (Ch. 3) and the multivariate CLT.

Chapter 3. The Multivariate Normal Distribution

Definition and properties, conditional distributions, best linear prediction, multiple and partial correlation, quadratic forms

1 Definition

1.1 Non-singular case

The univariate normal density is $f(x) = (2\pi\sigma^2)^{-1/2}\exp[-\frac{1}{2}(x-\mu)\sigma^{-2}(x-\mu)]$. Replacing $(x-\mu)^2/\sigma^2$ by a squared Mahalanobis distance gives the multivariate version.

Definition 3.1: multivariate normal (non-singular)

For $\boldsymbol{\mu} \in \mathbb{R}^p$ and $\boldsymbol{\Sigma}_{(p\times p)} \succ 0$, $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ means $\mathbf{X}$ has density

$$f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2}\exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big], \qquad \mathbf{x}\in\mathbb{R}^p,$$

where $|2\pi\boldsymbol{\Sigma}| = (2\pi)^p|\boldsymbol{\Sigma}|$.

$\int f = 1$. Let $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})$, so $d\mathbf{y}/d\mathbf{x} = |\boldsymbol{\Sigma}|^{-1/2}$. Then

$$\int f(\mathbf{x})\,d\mathbf{x} = \int\cdots\int (2\pi)^{-p/2}|\boldsymbol{\Sigma}|^{-1/2}e^{-\mathbf{y}^\top\mathbf{y}/2}|\boldsymbol{\Sigma}|^{1/2}d\mathbf{y} = \prod_{i=1}^{p}\int (2\pi)^{-1/2}e^{-y_i^2/2}dy_i = 1.$$

Equivalently, with $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$ and $\mathbf{z} = \mathbf{Q}^\top(\mathbf{x}-\boldsymbol{\mu})$, the exponent becomes $\sum_i z_i^2/\lambda_i$ and the integral is a product of univariate Gaussian integrals. ■

The substitution shows $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I}_p)$, i.e. $\mathbf{X} \overset{d}{=} \boldsymbol{\Sigma}^{1/2}\mathbf{Y} + \boldsymbol{\mu}$. Hence $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$, $\mathrm{Var}\mathbf{X} = \boldsymbol{\Sigma}$, and

$$U = (\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) = \mathbf{Y}^\top\mathbf{Y} = \textstyle\sum_i Y_i^2 \sim \chi^2(p).$$

1.2 General definition and Cramér–Wold

Definition 3.2: multivariate normal (singular allowed)

$\mathbf{X} \in \mathbb{R}^p$ has a multivariate normal distribution iff $\mathbf{t}^\top\mathbf{X}$ is univariate normal for every $\mathbf{t} \in \mathbb{R}^p$ (including the point mass $N(c, 0)$).

This definition does not require $\mathrm{Cov}(\mathbf{X})$ to be non-singular: $\mathbf{X}$ is MVN iff *every* projection onto a one-dimensional subspace is normal. It is well defined because, by Cramér–Wold (Ch. 2), the distributions of all $\mathbf{t}^\top\mathbf{X}$ determine that of $\mathbf{X}$.

Example 3.3: normal marginals do not imply joint normality

$X_1 \sim N(0,1)$ and $X_2 = -X_1$ if $-1 \le X_1 \le 1$, $X_2 = X_1$ otherwise.
(a) $X_2 \sim N(0,1)$.
(b) (X_1, X_2) is not bivariate normal.

Proof. (a) X_1 is symmetric about 0, so $-X_1 \overset{d}{=} X_1$ on $\{-1 \le X_1 \le 1\}$: $P[X_2 \le x_2] = P[-X_1 \le x_2, |X_1| \le 1] + P[X_1 \le x_2, |X_1| > 1] = P[X_1 \le x_2]$.

(b) $X_1 - X_2 = 2X_1$ if $|X_1| \le 1$ and 0 otherwise, so $P(X_1 - X_2 = 0) = P(|X_1| > 1) \approx 0.317 \in (0,1)$: $X_1 - X_2$ is neither a continuous normal nor a point mass. A linear combination is not normal, so (X_1, X_2) is not jointly normal. ■

1.3 Characteristic function

Theorem 3.4: characteristic function of the MVN

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succeq 0$, then $\varphi_{\mathbf{X}}(\mathbf{t}) = \exp\left(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t}\right)$. If $\boldsymbol{\Sigma} \succ 0$, the density above exists.

Proof. $\mathbf{t}^\top \mathbf{X} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t})$ and $\varphi_{N(m,v)}(s) = \exp(ism - \frac{1}{2}s^2 v)$, so $\varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top \mathbf{X}}(1) = \exp(i\mathbf{t}^\top \boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma}\mathbf{t})$. ■

Since the characteristic function depends only on $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the notation $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is justified under the general definition as well.

2 Geometry: Constant-Density Ellipsoids

Location parameter $\boldsymbol{\mu}$, shape parameter $\boldsymbol{\Sigma} \succ 0$. The contours of constant density are

$$E_c = \{\mathbf{x} \in \mathbb{R}^p : f(\mathbf{x}) = c_0\} = \{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2\},$$

where c is the Mahalanobis distance $d_M(\boldsymbol{\mu}, \mathbf{x})$. With $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\lambda_1 \ge \cdots \ge \lambda_p > 0$, E_c is an ellipsoid centered at $\boldsymbol{\mu}$ with principal axes along $\mathbf{u}_i$ and half-lengths proportional to $\sqrt{\lambda_i}$. If $\boldsymbol{\Sigma} = \mathbf{I}_p$ it is a sphere of radius c.

Example 3.5: bivariate normal

$\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{R}_{\pi/4} \mathrm{diag}(3,1) \mathbf{R}_{\pi/4}^\top$: elliptical contours with principal axes at $45°$ and $-45°$ and half-axis ratio $\sqrt{3} : 1$.

3 Linear Transformations, Subsets, Independence, Conditionals

Motivation. The multivariate normal is the default model because it is closed under every operation we perform: linear maps, marginals and conditionals are again normal, and zero correlation means independence. The conditional distribution formula is the population version of linear regression.

Theorem 3.6: linear combinations

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then $\mathbf{a}^\top \mathbf{X} = a_1 X_1 + \cdots + a_p X_p \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$. Conversely, if $\mathbf{a}^\top \mathbf{X} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a}$. Normality via the MGF: $M_{\mathbf{a}^\top \mathbf{X}}(t) = \mathrm{E}e^{t\mathbf{a}^\top \mathbf{X}} = M_{\mathbf{X}}(t\mathbf{a}) = \exp(t\mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2}t^2 \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$, the MGF of $N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma}\mathbf{a})$. The converse is the general definition. ■

Theorem 3.7: linear transformations

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{A} \in \mathbb{R}^{q\times p}$ and $\mathbf{d} \in \mathbb{R}^p$, then $\mathbf{AX} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$ and $\mathbf{X}+\mathbf{d} \sim N_p(\boldsymbol{\mu}+\mathbf{d}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{AX}) = \mathbf{A}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{AX}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. Any $\mathbf{b}^\top(\mathbf{AX}) = (\mathbf{A}^\top\mathbf{b})^\top\mathbf{X}$ is a linear combination of $\mathbf{X}$, normal by Theorem 3.6, so $\mathbf{AX}$ is MVN. $\mathbf{a}^\top(\mathbf{X}+\mathbf{d}) = \mathbf{a}^\top\mathbf{X} + \mathbf{a}^\top\mathbf{d}$ only shifts the mean. ■

Theorem 3.8: subsets

Partition $\mathbf{X} = \begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix}$, $\boldsymbol{\mu} = \begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}$. Then $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$; all subsets of $\mathbf{X}$ are normal.

Proof. Theorem 3.7 with $\mathbf{A} = [\mathbf{I}_{q\times q}\ \mathbf{0}_{q\times(p-q)}]$. Any subset can be relabelled as $\mathbf{X}_1$. ■

Theorem 3.9: independence and covariance

(a) If $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$, then $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$.

(b) If $\begin{pmatrix}\mathbf{X}_1\\ \mathbf{X}_2\end{pmatrix} \sim N_{q_1+q_2}$, then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2 \iff \boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) If $\mathbf{X}_1 \sim N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ are independent, then $(\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_{q_1+q_2}\Big(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}\Big)$.

Proof. (b) The joint characteristic function is

$$\varphi_{\mathbf{X}}(\mathbf{t}_1, \mathbf{t}_2) = \exp\Big\{ i(\mathbf{t}_1^\top\boldsymbol{\mu}_1 + \mathbf{t}_2^\top\boldsymbol{\mu}_2) - \tfrac{1}{2}\begin{pmatrix}\mathbf{t}_1\\ \mathbf{t}_2\end{pmatrix}^\top \begin{pmatrix}\boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12}\\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22}\end{pmatrix}\begin{pmatrix}\mathbf{t}_1\\ \mathbf{t}_2\end{pmatrix} \Big\}.$$

If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ it factorizes as $\varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$: independence. Conversely, independence forces the factorization, so the cross term $\mathbf{t}_1^\top\boldsymbol{\Sigma}_{12}\mathbf{t}_2$ must vanish for all $\mathbf{t}$, i.e. $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) reads the factorization backwards. ■

Remark 3.10: "uncorrelated ⇒ independent" needs joint normality

In Example 3.3 each marginal is normal but the pair is not jointly normal, so (b) does not apply.

Theorem 3.11: conditional distribution

If $\boldsymbol{\Sigma}_{22} \succ 0$,

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \ \sim\ N_r\big(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big).$$

The conditional mean is linear in $\mathbf{x}_2$; the conditional covariance $\boldsymbol{\Sigma}_{11\cdot 2} := \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ does not depend on $\mathbf{x}_2$.

Proof. Define $\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, $\mathbf{X}_2^* = \mathbf{X}_2$, i.e. $\mathbf{X}^* = \mathbf{AX}$ with $\mathbf{A} = \begin{pmatrix}\mathbf{I}_r & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\\ \mathbf{0} & \mathbf{I}_s\end{pmatrix}$. By Theorem 3.7 $\mathbf{X}^*$ is MVN and

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \begin{pmatrix}\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{0}\\ \mathbf{0} & \boldsymbol{\Sigma}_{22}\end{pmatrix}.$$

$\mathrm{Cov}(\mathbf{X}_1^*, \mathbf{X}_2^*) = \mathbf{0}$, so $\mathbf{X}_1^* \perp\!\!\!\perp \mathbf{X}_2$ (Theorem 3.9(b)) and $\mathbf{X}_1^* \sim N_r(\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{11\cdot 2})$. Since $\mathbf{X}_1 = \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2 = \mathbf{x}_2$ makes the second term constant without changing the law of $\mathbf{X}_1^*$: $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \overset{d}{=} \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2$. ■

Remark 3.12: residuals, projections and partial variance

$\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ is (up to centring) the residual $\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)$ after projecting $\mathbf{X}_1$ onto the span of $\mathbf{X}_2$. The residual is uncorrelated with $\mathbf{X}_2$, and independent of it in the Gaussian case. $\sigma_{11\cdot 2}^2 = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$ (more generally $\boldsymbol{\Sigma}_{11\cdot 2}$) is the **partial variance** of $\mathbf{X}_1$ after removing the effect of $\mathbf{X}_2$, i.e. $\mathrm{Var}(\mathbf{X}_1^*)$.

4 Best Linear Prediction, Multiple and Partial Correlation

Motivation. How well can Y be predicted linearly from $\mathbf{X}$, and how correlated are two variables once a third is held fixed? Both answers come from the conditional covariance of the normal distribution.

Let $\mathbf{X} \in \mathbb{R}^q$ be the predictor and $Y \in \mathbb{R}$ the response, $\mathbf{U} = (\mathbf{X}^\top, Y)^\top \in \mathbb{R}^p$ $(p = q + 1)$,

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\mu}_X \\ \mu_Y \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \mathrm{Var}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\Sigma}_{XX} & \boldsymbol{\Sigma}_{XY} \\ \boldsymbol{\Sigma}_{YX} & \sigma_Y^2 \end{pmatrix},$$

with sample counterparts $\bar{\mathbf{u}} = (\bar{\mathbf{x}}^\top, \bar{y})^\top$ and $\mathbf{S} = \begin{pmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & s_Y^2 \end{pmatrix}$.

4.1 Best linear predictor

Theorem 3.13: best linear predictor

$\mathrm{BLP}(Y \mid \mathbf{X}) := \mathbf{A}^*\mathbf{X} + b^*$, $(\mathbf{A}^*, b^*) = \arg\min_{\mathbf{A} \in \mathbb{R}^{1\times q},\, b \in \mathbb{R}} \mathrm{E}\{Y - (\mathbf{A}\mathbf{X} + b)\}^2$. With finite second moments and $\boldsymbol{\Sigma}_{XX} \succ 0$,

$$\mathbf{A}^* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}, \qquad b^* = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X, \qquad \mathrm{BLP}(Y \mid \mathbf{X}) = \mu_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X).$$

Proof. Let $W := Y - \mu_Y - \mathbf{A}^*(\mathbf{X} - \boldsymbol{\mu}_X)$; then $\mathrm{E}W = 0$ and $\mathrm{Cov}(W, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{A}^*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$. For any $(\mathbf{A}, b)$,

$$Y - \mathbf{A}\mathbf{X} - b = W + (\mathbf{A}^* - \mathbf{A})(\mathbf{X} - \boldsymbol{\mu}_X) + \{\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b\},$$

three mutually uncorrelated terms (the last is constant), so

$$\mathrm{E}(Y - \mathbf{A}\mathbf{X} - b)^2 = \mathrm{E}W^2 + (\mathbf{A}^* - \mathbf{A})\boldsymbol{\Sigma}_{XX}(\mathbf{A}^* - \mathbf{A})^\top + (\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b)^2 \geq \mathrm{E}W^2,$$

with equality iff $\mathbf{A} = \mathbf{A}^*$ and $b = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X$. ■

Remark 3.14: properties of the BLP

$\mathrm{E}[\mathrm{BLP}(Y \mid \mathbf{X})] = \mu_Y$, $\mathrm{Var}[\mathrm{BLP}(Y \mid \mathbf{X})] = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$; the residual $W = Y - \mathrm{BLP}(Y \mid \mathbf{X})$ has $\mathrm{E}W = 0$, $\mathrm{Var}W = \sigma_Y^2 - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\mathrm{Cov}(W, \mathrm{BLP}) = 0$. Under joint normality $\mathrm{BLP}(Y \mid \mathbf{X}) = \mathrm{E}(Y \mid \mathbf{X})$ (Theorem 3.11). In block notation $\mathrm{E}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\mu}_2$, $\mathrm{Var}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$.

4.2 Multiple correlation

Definition 3.15: multiple correlation of Y with $\mathbf{X}$

$$\rho_{\text{mult}} = \text{mcorr}(\mathbf{X}, Y) := \max_{\mathbf{a}\in\mathbb{R}^q\setminus\{\mathbf{0}\}} \text{Corr}(Y, \mathbf{a}^\top\mathbf{X}), \qquad \rho^2_{\text{mult}} = \frac{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}{\sigma_Y^2}.$$

With $\hat{Y} = \text{BLP}(Y \mid \mathbf{X})$, $\rho_{\text{mult}} = \text{Corr}(Y, \hat{Y})$.

Proof. $\text{Corr}(Y, \mathbf{a}^\top\mathbf{X} + b) = \dfrac{\mathbf{a}^\top\boldsymbol{\Sigma}_{XY}}{\sigma_Y\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}_{XX}\mathbf{a}}}$. With $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{1/2}\mathbf{a}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top\boldsymbol{\Sigma}_{XY} = \mathbf{b}^\top\boldsymbol{\Sigma}_{XX}^{-1/2}\boldsymbol{\Sigma}_{XY} \le \sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}_{XX}\mathbf{a}}\,\sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}},$$

so Corr $\le \sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}/\sigma_Y$, with equality iff $\mathbf{a} \propto \boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ (the BLP direction). Then $\text{Cov}(Y, \hat{Y}) = \text{Var}(\hat{Y}) = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\text{Corr}(Y, \hat{Y}) = \sqrt{\text{Var}\hat{Y}}/\sigma_Y$. ∎

For two vectors, $\rho(\mathbf{X}_1, \mathbf{X}_2) := \max \text{Corr}(\mathbf{a}^\top\mathbf{X}_1, \mathbf{b}^\top\mathbf{X}_2)$ is the largest canonical correlation (Ch. 8).

4.3 Partial covariance and partial correlation

Partition $\mathbf{X} = (X, \mathbf{Z}^\top)^\top$ with X the predictor of interest and $\mathbf{Z} \in \mathbb{R}^{q-1}$ the others ($\text{Var}\mathbf{Z} \succ 0$), and let $\tilde{X} = X - \text{BLP}(X \mid \mathbf{Z})$, $\tilde{Y} = Y - \text{BLP}(Y \mid \mathbf{Z})$.

Definition 3.16: partial covariance and partial correlation

$$\sigma_{YX\cdot Z} := \text{Cov}(\tilde{Y}, \tilde{X}) = \text{Cov}(Y, X) - \text{Cov}(Y, \mathbf{Z})\text{Var}(\mathbf{Z})^{-1}\text{Cov}(\mathbf{Z}, X),$$

$$\rho_{YX\cdot Z} := \frac{\sigma_{YX\cdot Z}}{\sqrt{\text{Var}(\tilde{Y})\text{Var}(\tilde{X})}} = \text{Corr}(\tilde{Y}, \tilde{X}).$$

In block notation: $\text{pcorr}(\mathbf{X}_1, \mathbf{X}_2; \mathbf{X}_3) = \text{Corr}(\mathbf{X}_1 - \text{BLP}(\mathbf{X}_1 \mid \mathbf{X}_3), \mathbf{X}_2 - \text{BLP}(\mathbf{X}_2 \mid \mathbf{X}_3)) = \text{Corr}(\mathbf{X}_1^*, \mathbf{X}_2^*)$ and the partial covariance matrix $\text{pVar}(\mathbf{X}_1; \mathbf{X}_2) = \text{Var}(\mathbf{X}_1 - \text{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$.

Partial covariance formula. $\tilde{Y} = Y - \mu_Y - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}(\mathbf{Z} - \boldsymbol{\mu}_Z)$ and similarly for $\tilde{X}$. Expanding, $\text{Cov}(\tilde{Y}, \tilde{X}) = \sigma_{YX} - 2\boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} + \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} = \sigma_{YX} - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX}$. ∎

So the partial covariance matrix of (Y, X) is a Schur complement, and under joint normality it equals the conditional covariance given $\mathbf{Z}$.

Theorem 3.17: Frisch–Waugh–Lovell theorem

Let $\hat{\beta}_X$ be the coefficient of X in the OLS regression of Y on $(X, \mathbf{Z})$, and let $\tilde{y}_i, \tilde{x}_i$ be the residuals from the OLS regressions of Y and X on $\mathbf{Z}$. Then

$$\hat{\beta}_X = \frac{\sum_i \tilde{x}_i\tilde{y}_i}{\sum_i \tilde{x}_i^2},$$

the slope of the residual-on-residual regression (all regressions with intercept; full-column-rank design).

Proof. Let $\mathbf{M}_Z = \mathbf{I} - \mathbf{P}_{[\mathbf{1},\mathbf{Z}]}$. The normal equations give $\mathbf{y} = \hat\beta_X \mathbf{x} + [\mathbf{1}, \mathbf{Z}]\hat\gamma + \mathbf{e}$ with $\mathbf{e} \perp \mathbf{x}, \mathbf{1}, \mathbf{Z}$. Multiplying by $\mathbf{M}_Z$ ($\mathbf{M}_Z[\mathbf{1}, \mathbf{Z}] = \mathbf{0}$, $\mathbf{M}_Z\mathbf{e} = \mathbf{e}$) gives $\tilde{\mathbf{y}} = \hat\beta_X \tilde{\mathbf{x}} + \mathbf{e}$. Multiply by $\tilde{\mathbf{x}}^\top$: $\tilde{\mathbf{x}}^\top\mathbf{e} = \mathbf{x}^\top\mathbf{M}_Z\mathbf{e} = \mathbf{x}^\top\mathbf{e} = 0$, so $\tilde{\mathbf{x}}^\top\tilde{\mathbf{y}} = \hat\beta_X \tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}$. ■

Example 3.18: mtcars — marginal versus partial correlation

Y = mpg, X = displacement, $\mathbf{Z}$ = (horsepower, weight). The marginal correlation is $\hat\rho_{YX} \approx -0.805$, but after adjusting for horsepower and weight $\hat\rho_{YX\cdot Z} \approx -0.044$: little linear association remains. The displacement–mpg relation runs mostly through horsepower and weight.

5 Quadratic Forms in Normal Random Vectors

Motivation. Test statistics are typically quadratic forms such as $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})$. Their χ^2 distributions are the basis of all tests in later chapters.

For $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{A}$, a quadratic form is $Y = \mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j,k} X_j a_{jk} X_k$. We want its distribution and conditions for independence. Special case: $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, $\mathbf{A} = \mathbf{I}_p$ gives $Y = \sum X_i^2 \sim \chi^2(p)$.

Remark 3.19: matrix algebra recall

$\mathbf{A}$ is idempotent if $\mathbf{A}^2 = \mathbf{A}$. If $\mathbf{A}$ is symmetric, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$; if it is also idempotent, its eigenvalues are 0 or 1 and $\operatorname{rank}(\mathbf{A}) = \#\{\text{nonzero eigenvalues}\} = \operatorname{tr}(\mathbf{A})$.

Theorem 3.20: chi-square quadratic forms, $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$

Let $\mathbf{X} \sim N_p(\mathbf{0}, \sigma^2\mathbf{I})$ and $\mathbf{A}$ symmetric. Then $Y = \dfrac{\mathbf{X}^\top\mathbf{A}\mathbf{X}}{\sigma^2} \sim \chi^2(m)$ iff $\mathbf{A}$ is idempotent of rank $m \le p$.

Proof. ($\Leftarrow$) $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \operatorname{diag}(1, \dots, 1, 0, \dots, 0)$ (m ones). Let $\mathbf{z} = \frac{1}{\sigma}\boldsymbol{\Gamma}^\top\mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Gamma}^\top\boldsymbol{\Gamma}) = N(\mathbf{0}, \mathbf{I}_p)$; then $Y = \mathbf{z}^\top\boldsymbol{\Lambda}\mathbf{z} = z_1^2 + \cdots + z_m^2 \sim \chi^2(m)$.

($\Rightarrow$) In general $Y = \sum_i \lambda_i z_i^2$ has MGF $\prod_i (1 - 2t\lambda_i)^{-1/2}$; matching $(1 - 2t)^{-m/2}$ near $t = 0$ forces the nonzero λ_i to be m ones. ■

Simple example: $\mathbf{A} = \operatorname{diag}(1, 0, \dots, 0)$ gives $\mathbf{X}^\top\mathbf{A}\mathbf{X}/\sigma^2 = X_1^2/\sigma^2 \sim \chi_1^2$.

Theorem 3.21: chi-square quadratic forms, general $\boldsymbol{\Sigma}$

Let $\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $\mathbf{A}$ symmetric. Then $Y = \mathbf{X}^\top\mathbf{A}\mathbf{X} \sim \chi^2(m)$ iff either (1) $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent of rank m, or (2) $\boldsymbol{\Sigma}\mathbf{A}$ is idempotent of rank m.

Proof. By the representation theorem $\mathbf{X} = \boldsymbol{\Sigma}^{1/2}\mathbf{z}$, $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so $Y = \mathbf{z}^\top\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}\mathbf{z}$. By Theorem 3.20, $Y \sim \chi_m^2$ iff $\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}$ is (symmetric) idempotent of rank m. Now

$$(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2})^2 = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} \iff \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} \iff \mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A},$$

(multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides), and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A} \iff (\mathbf{A}\boldsymbol{\Sigma})^2 = \mathbf{A}\boldsymbol{\Sigma}$ (multiply on the right by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$). Ranks: $\operatorname{rank}(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2}) = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\boldsymbol{\Sigma})$. Since $\boldsymbol{\Sigma}\mathbf{A} = (\mathbf{A}\boldsymbol{\Sigma})^\top$, the same holds for $\boldsymbol{\Sigma}\mathbf{A}$. ■

Example 3.22: distribution of the Mahalanobis distance

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, then $(\mathbf{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi^2(p)$ ($\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ makes $\mathbf{A}\boldsymbol{\Sigma} = \mathbf{I}_p$ idempotent of rank p). Hence $\{\mathbf{x} : (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \le \chi^2_p(\alpha)\}$ is an ellipsoid of probability $1-\alpha$.

Theorem 3.23: independence of linear and quadratic forms

Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, $\mathbf{A}$ a $p \times p$ symmetric matrix and $\mathbf{B}$ a $k \times p$ matrix. If $\mathbf{BA} = \mathbf{0}$, then $\mathbf{BX}$ and $\mathbf{X}^\top \mathbf{AX}$ are independent.

Proof. If $\mathbf{A}$ is full rank, $\mathbf{BA} = \mathbf{0}$ forces $\mathbf{B} = \mathbf{0}$: trivial. If $\operatorname{rank}(\mathbf{A}) = m < p$, write $\mathbf{A} = \boldsymbol{\Gamma} \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda}_m = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, all $\lambda_i \neq 0$. From $\mathbf{BA} = \mathbf{0}$, $\mathbf{B}\boldsymbol{\Gamma} \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} = \mathbf{0}$. Writing $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{C}_1, \mathbf{C}_2]$ gives $\mathbf{C}_1 \boldsymbol{\Lambda}_m = \mathbf{0}$, so $\mathbf{C}_1 = \mathbf{0}$ and $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{0}, \mathbf{C}_2]$. Let $\mathbf{z} = \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix} = \boldsymbol{\Gamma}^\top \mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1 \perp\!\!\!\perp \mathbf{z}_2$. Then

$$\mathbf{BX} = \mathbf{B}\boldsymbol{\Gamma}\mathbf{z} = \mathbf{C}_2 \mathbf{z}_2, \qquad \mathbf{X}^\top \mathbf{AX} = \mathbf{z}^\top \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{z} = \mathbf{z}_1^\top \boldsymbol{\Lambda}_m \mathbf{z}_1,$$

functions of $\mathbf{z}_2$ and $\mathbf{z}_1$ respectively, hence independent. ■

Remark 3.24: same argument, two quadratic forms

If $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and symmetric $\mathbf{A}, \mathbf{B}$ satisfy $\mathbf{AB} = \mathbf{0}$, then $\mathbf{X}^\top \mathbf{AX} \perp\!\!\!\perp \mathbf{X}^\top \mathbf{BX}$. Independence of $\bar{X}$ and S^2 is the case $\mathbf{B} = \frac{1}{n}\mathbf{1}^\top$, $\mathbf{A} = \mathbf{H}_n$ with $\mathbf{BA} = \mathbf{0}$; the multivariate version is in Ch. 4.

Chapter 4. Sampling Distributions and Large-Sample Theory

The Wishart distribution, distributions of $\bar{\mathbf{X}}$ and $\mathbf{S}$, MLE, consistency, multivariate CLT

1 The Wishart Distribution

Motivation. The sample covariance matrix is random, and inference about $\boldsymbol{\Sigma}$ needs its distribution. The Wishart distribution is the matrix analogue of χ^2: a sum of outer products of independent normal vectors.

Definition 4.1: Wishart distribution

- The Wishart distribution is a family of distributions for symmetric positive definite matrices.
- The distribution of the $p \times p$ random matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{X} = \sum_{i=1}^{n} \mathbf{X}_i \mathbf{X}_i^\top$ is a Wishart distribution.
- With m degrees of freedom,

$$W_p(m \mid \boldsymbol{\Sigma}) = \text{Wishart with } m \text{ d.f.} = \text{distribution of } \sum_{j=1}^{m} \mathbf{Z}_j \mathbf{Z}_j^\top, \qquad \mathbf{Z}_j \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

We also write $W_p(m, \boldsymbol{\Sigma})$.

- It is the multivariate extension of χ^2: if $\mathbf{M} \sim W_1(n, \sigma^2)$ then $\mathbf{M}/\sigma^2 \sim \chi^2_n$. E.g. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0, \sigma^2)$ gives $\mathbf{M} = \sum X_i^2$, $\mathbf{M}/\sigma^2 \sim \chi^2_n$.
- $W_p(n, \mathbf{I})$ is the *standard Wishart* distribution.
- If $\boldsymbol{\mu} = \mathbf{0}$ is known and $\mathbf{S} = \frac{1}{n}\mathbf{X}^\top \mathbf{X} = \frac{1}{n}\sum \mathbf{X}_i \mathbf{X}_i^\top$, then $n\mathbf{S} \sim W_p(n, \boldsymbol{\Sigma})$.
- If $m \geq p$ and $\boldsymbol{\Sigma} \succ 0$, $\mathbf{M}$ is PD with probability 1.

Theorem 4.2: properties of the Wishart distribution

(i) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ and $\mathbf{B}_{(p \times m)}$, $\mathbf{B}^\top \mathbf{M} \mathbf{B} \sim W_m(n, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) For $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, $\boldsymbol{\Sigma}^{-1/2} \mathbf{M} \boldsymbol{\Sigma}^{-1/2} \sim W_p(n, \mathbf{I}_p)$.

(iii) If $\mathbf{M}_i \sim W_p(n_i, \boldsymbol{\Sigma})$ $(i = 1, \ldots, k)$ are independent, $\sum_i \mathbf{M}_i \sim W_p(n, \boldsymbol{\Sigma})$ with $n = n_1 + \cdots + n_k$.

(iv) For $\mathbf{M}_n \sim W_p(n, \boldsymbol{\Sigma})$, $\mathrm{E}(\mathbf{M}_n) = n\boldsymbol{\Sigma}$ and $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$. Hence $\hat{\boldsymbol{\Sigma}}_n = \mathbf{M}_n/n$ has $\mathrm{E}(\hat{\boldsymbol{\Sigma}}_n) = \boldsymbol{\Sigma}$ and $\mathrm{Var}[(\hat{\boldsymbol{\Sigma}}_n)_{ij}] = (\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/n$.

(v) The law of large numbers and the Cramér–Wold device give $\mathbf{M}_n/n \to \boldsymbol{\Sigma}$ in probability as $n \to \infty$.

Proof. (i) Write $\mathbf{M} = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Then $\mathbf{B}^\top \mathbf{M} \mathbf{B} = \sum_i \mathbf{B}^\top \mathbf{x}_i (\mathbf{B}^\top \mathbf{x}_i)^\top$ and $\mathbf{B}^\top \mathbf{x}_i \overset{\text{iid}}{\sim} N_m(\mathbf{0}, \mathbf{B}^\top \boldsymbol{\Sigma} \mathbf{B})$.

(ii) Special case of (i) with $\mathbf{B} = \boldsymbol{\Sigma}^{-1/2}$.

(iii) With $\mathbf{M}_i = \sum_j \mathbf{x}_{ij}\mathbf{x}_{ij}^\top$, $\sum_i \mathbf{M}_i = \sum_i \sum_j \mathbf{x}_{ij}\mathbf{x}_{ij}^\top \sim W_p(\sum n_i, \boldsymbol{\Sigma})$.

(iv) $\mathrm{E}[\mathbf{M}_n] = \mathrm{E}(\sum_k \mathbf{x}_k\mathbf{x}_k^\top) = \sum_k \mathrm{Var}(\mathbf{x}_k) = n\boldsymbol{\Sigma}$. Terms with different k are independent, so $\mathrm{Var}((\mathbf{M}_n)_{ij}) = n\{\mathrm{E}(X_{ki}^2X_{kj}^2) - \Sigma_{ij}^2\}$. Using $\mathrm{E}(X_{ki} \mid X_{kj}) = \frac{\Sigma_{ij}}{\Sigma_{jj}}X_{kj}$ and $X_{ki} - \mathrm{E}(X_{ki} \mid X_{kj}) \perp\!\!\!\perp X_{kj}$,

$$\mathrm{E}[(X_{ki}X_{kj})^2] = \mathrm{E}[X_{kj}^2\mathrm{E}(X_{ki}^2 \mid X_{kj})] = \tfrac{\Sigma_{ij}^2}{\Sigma_{jj}^2}3\Sigma_{jj}^2 + \Big(\Sigma_{ii} - \tfrac{\Sigma_{ij}^2}{\Sigma_{jj}}\Big)\Sigma_{jj} = 2\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj},$$

so $\mathrm{Var} = n(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})$.

(v) By Cramér–Wold it suffices that $\sum_{ij} a_{ij}M_{ij}/n \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$ for all a_{ij}. Since $\mathbf{M}_n/n = \frac{1}{n}\sum_k \mathbf{x}_k\mathbf{x}_k^\top$ is an average of iid random matrices, the LLN gives $\sum_{ij} a_{ij}\frac{M_{ij}}{n} \xrightarrow{p} \sum_{ij} a_{ij}\Sigma_{ij}$. ■

2 Wishart Quadratic Forms and the Distribution of S

Motivation. Projections of a Wishart matrix are again Wishart or χ^2, which yields the distribution of $\mathbf{S}$ and its independence from $\bar{\mathbf{X}}$, just as in the univariate case.

Theorem 4.3: Wishart quadratic forms

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top$, and $\mathbf{A}$ an $n \times n$ real symmetric matrix. Then

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} \sim W_p(r, \boldsymbol{\Sigma}) \iff \mathbf{A} \text{ is idempotent with } \mathrm{rank}(\mathbf{A}) = r.$$

Partial proof ($\Leftarrow$). Since $\mathbf{A}$ is symmetric idempotent of rank r, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \mathrm{diag}(1, \dots, 1, 0, \dots, 0)$. Write $\mathbf{A} = \mathbf{P}\begin{pmatrix}\mathbf{I}_r & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{pmatrix}\mathbf{P}^\top$ for an $n \times n$ orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$, so $\mathbf{A} = \sum_{j=1}^r \mathbf{p}_j\mathbf{p}_j^\top$ and

$$\mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j=1}^r \mathbf{X}^\top\mathbf{p}_j\mathbf{p}_j^\top\mathbf{X}.$$

Now $\mathbf{X}^\top\mathbf{p}_j = \sum_{i=1}^n p_{ij}\mathbf{X}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, since

$$\mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i\Big) = \boldsymbol{\Sigma}\cdot\mathbf{p}_j^\top\mathbf{p}_j = \boldsymbol{\Sigma}, \qquad \mathrm{Cov}\Big(\sum_i p_{ij}\mathbf{X}_i, \sum_i p_{ik}\mathbf{X}_i\Big) = \mathbf{p}_j^\top\mathbf{p}_k\boldsymbol{\Sigma} = \mathbf{0}\ (j \neq k);$$

being jointly normal, $\mathbf{z}_j = \mathbf{X}^\top\mathbf{p}_j$ are iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$. Hence $\mathbf{X}^\top\mathbf{A}\mathbf{X} = \sum_{j=1}^r \mathbf{z}_j\mathbf{z}_j^\top \sim W_p(r, \boldsymbol{\Sigma})$. ($\Rightarrow$ reduces to Theorem 3.20 with $p = 1$: $\mathbf{a}^\top\mathbf{X}^\top\mathbf{A}\mathbf{X}\mathbf{a} \sim (\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a})\chi_r^2$ for every $\mathbf{a}$.) ■

Theorem 4.4: distribution of the sample covariance matrix; $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$

If $\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

$$\bar{\mathbf{X}}_n \sim N_p(\boldsymbol{\mu}, \tfrac{1}{n}\boldsymbol{\Sigma}), \qquad (n-1)\mathbf{S} = \mathbf{X}^\top\Big(\mathbf{I}_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\Big)\mathbf{X} \sim W_p(n-1, \boldsymbol{\Sigma}), \qquad \bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}.$$

Proof.

Step 1. $\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ is symmetric idempotent of rank $n-1$. Since it annihilates $\mathbf{1}_n$, $\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = (\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)^\top\mathbf{H}_n(\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)$, which reduces to the mean-zero case, and Theorem 4.3 gives $W_p(n-1, \boldsymbol{\Sigma})$. Independence: choose an orthogonal $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ with $\mathbf{p}_1 = \mathbf{1}_n/\sqrt{n}$ and let $\mathbf{z}_j = (\mathbf{X} - \mathbf{1}_n\boldsymbol{\mu}^\top)^\top\mathbf{p}_j$. As above $\mathbf{z}_1, \dots, \mathbf{z}_n \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\mathbf{z}_1 = \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})$, and since $\mathbf{H}_n = \sum_{j=2}^n \mathbf{p}_j\mathbf{p}_j^\top$, $(n-1)\mathbf{S} = \sum_{j=2}^n \mathbf{z}_j\mathbf{z}_j^\top$. The first depends only on $\mathbf{z}_1$, the second only on $\mathbf{z}_2, \dots, \mathbf{z}_n$. ■

3 A Theorem on Inverse Wishart Ratios

Motivation. Hotelling's T^2 involves $\mathbf{S}^{-1}$. This theorem reduces quadratic forms in an inverse Wishart matrix to ratios of χ^2 variables, which is the key step toward the F distribution of T^2.

Theorem 4.5: ratios involving an inverse Wishart matrix

Suppose $\mathbf{M} \sim W_p(n, \boldsymbol{\Sigma})$, $\mathbf{a} \in \mathbb{R}^p$, and $n > p-1$. Then

(i) $(\boldsymbol{\Sigma}^{-1})_{11}/(\mathbf{M}^{-1})_{11} \sim \chi^2_{n-p+1}$;

(ii) for any constant $\mathbf{a} \neq \mathbf{0}$, $\dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top \mathbf{M}^{-1}\mathbf{a}} \sim \chi^2_{n-p+1}$;

(iii) if $\mathbf{y} \in \mathbb{R}^p$ is random and independent of $\mathbf{M}$ (with $P(\mathbf{y}=\mathbf{0})=0$), then $\dfrac{\mathbf{y}^\top \boldsymbol{\Sigma}^{-1}\mathbf{y}}{\mathbf{y}^\top \mathbf{M}^{-1}\mathbf{y}} \sim \chi^2_{n-p+1}$ and is independent of $\mathbf{y}$.

Proof. (i) Write

$$\mathbf{M} = \begin{pmatrix} m_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

By the block-inverse formula,

$$\frac{(\boldsymbol{\Sigma}^{-1})_{11}}{(\mathbf{M}^{-1})_{11}} = \frac{m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21}}{\sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}}.$$

Write $\mathbf{M} = \mathbf{X}^\top\mathbf{X} = \sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^\top$ with $\mathbf{x}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$ ($\mathbf{x}_1$: first column). Then

$$m_{11} - \mathbf{M}_{12}\mathbf{M}_{22}^{-1}\mathbf{M}_{21} = \mathbf{x}_1^\top\mathbf{x}_1 - \mathbf{x}_1^\top\mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{x}_1 = \mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1,$$

$$\mathbf{A} = \mathbf{A}(\mathbf{X}_2) = \mathbf{I}_n - \mathbf{X}_2(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top.$$

Given $\mathbf{X}_2$, $\mathbf{x}_1 \mid \mathbf{X}_2 \sim N_n(\mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21},\ \sigma_{11\cdot 2}\mathbf{I}_n)$, $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$. Since $\mathbf{A}\mathbf{X}_2 = \mathbf{0}$, $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 = (\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})^\top\mathbf{A}(\mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21})$. For any given $\mathbf{X}_2$, $\mathbf{A}$ is symmetric idempotent with $\operatorname{rank}(\mathbf{A}) = \operatorname{tr}(\mathbf{A}) = n - \operatorname{tr}\{(\mathbf{X}_2^\top\mathbf{X}_2)^{-1}\mathbf{X}_2^\top\mathbf{X}_2\} = n-(p-1)$, so Theorem 3.20 gives

$$\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \mid \mathbf{X}_2 \sim \sigma_{11\cdot 2}\,\chi^2_{n-p+1}.$$

This does not depend on $\mathbf{X}_2$, so marginally $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1 \sim \sigma_{11\cdot 2}\chi^2_{n-p+1}$, independent of $\mathbf{X}_2$. Hence the ratio is $\mathbf{x}_1^\top\mathbf{A}\mathbf{x}_1/\sigma_{11\cdot 2} \sim \chi^2_{n-p+1}$.

(i)⇒(ii) Given $\mathbf{a}$, take a $p\times p$ orthogonal $\mathbf{P}$ whose first column is $\mathbf{a}/\|\mathbf{a}\|$, i.e. $\mathbf{a}^\top\mathbf{P} = \mathbf{e}_1^\top$ (w.l.o.g. $\|\mathbf{a}\| = 1$). Then

$$\frac{\mathbf{a}^\top\boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}} = \frac{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}}{\mathbf{a}^\top\mathbf{P}\mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}\mathbf{P}^\top\mathbf{a}} = \frac{\mathbf{e}_1^\top\boldsymbol{\Sigma}_{\mathbf{P}}^{-1}\mathbf{e}_1}{\mathbf{e}_1^\top\mathbf{M}_{\mathbf{P}}^{-1}\mathbf{e}_1}, \quad \boldsymbol{\Sigma}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\boldsymbol{\Sigma}^{-1}\mathbf{P},\ \mathbf{M}_{\mathbf{P}}^{-1} = \mathbf{P}^\top\mathbf{M}^{-1}\mathbf{P}.$$

By Theorem 4.2, $\mathbf{M}_{\mathbf{P}} = \mathbf{P}^\top\mathbf{M}\mathbf{P} \sim W_p(n, \boldsymbol{\Sigma}_{\mathbf{P}})$, so (i) applies: $(\boldsymbol{\Sigma}_{\mathbf{P}}^{-1})_{11}/(\mathbf{M}_{\mathbf{P}}^{-1})_{11} \sim \chi^2_{n-p+1}$.

(ii)⇒(iii) Conditioning on $\mathbf{y} = \mathbf{a}$ leaves the law of $\mathbf{M}$ unchanged (independence), so by (ii) the conditional distribution is χ^2_{n-p+1} for almost all $\mathbf{a} \neq \mathbf{0}$. It does not depend on $\mathbf{a}$, so it is also the marginal law and is independent of $\mathbf{y}$. ∎

Remark 4.6: residuals and projections in multivariate analysis

1. The residual of $\mathbf{x}_1$ after projecting onto the span of $\mathbf{X}_2$ is $\mathbf{x}_1^* = \mathbf{x}_1 - \mathbf{X}_2\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathbf{x}_1 - \mathrm{BLP}(\mathbf{x}_1 \mid \mathbf{X}_2)$.
2. $\mathbf{x}_1^*$ is orthogonal to (uncorrelated with) $\mathbf{X}_2$; being Gaussian, they are independent.
3. The partial variance $\sigma_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21} = \mathrm{Var}(\mathbf{x}_1^*)$ is the variance left after removing $\mathbf{X}_2$.

The residual sum of squares has $n-(p-1)$ degrees of freedom. This theorem is the key to the F distribution of Hotelling's T^2 (Ch. 5).

4 Maximum Likelihood Estimation

Motivation. Under normality the maximum likelihood estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are the sample mean and the (biased) sample covariance. The proof uses a trace inequality rather than differentiation.

Theorem 4.7: MLE of μ and Σ

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be iid $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. The MLEs are

$$\hat{\boldsymbol{\mu}}_{\mathrm{MLE}} = \bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i, \qquad \hat{\boldsymbol{\Sigma}}_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top =: \mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}.$$

The MLE jointly minimizes the negative log-likelihood

$$\begin{aligned}\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \mathbf{x}^n) &:= -\log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) + c \\ &= \frac{n}{2}\log|\boldsymbol{\Sigma}| + \frac{n}{2}(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) + \frac{1}{2}\sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) + c.\end{aligned}$$

(Expand $\mathbf{x}_i - \boldsymbol{\mu} = (\mathbf{x}_i - \bar{\mathbf{x}}) + (\bar{\mathbf{x}} - \boldsymbol{\mu})$; the cross term vanishes.)

Step 1 (profile out $\boldsymbol{\mu}$). ℓ depends on $\boldsymbol{\mu}$ only through $(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \geq 0$, minimized by $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every $\boldsymbol{\Sigma}$. With $\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathrm{tr}\{\boldsymbol{\Sigma}^{-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\} = n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$,

$$2\ell(\hat{\boldsymbol{\mu}}_{\mathrm{MLE}}, \boldsymbol{\Sigma}) = n\log|\boldsymbol{\Sigma}| + n\mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) + c,$$

so it remains to minimize $f(\boldsymbol{\Sigma}) = \log|\boldsymbol{\Sigma}| + \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0)$ over $\boldsymbol{\Sigma} \succ 0$.

First proof: von Neumann's inequality. Decompose $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ and $\mathbf{S}_0 = \mathbf{V}\mathbf{D}\mathbf{V}^\top$ (λ_i, d_i in decreasing order; $d_i > 0$ a.s. when $n > p$). The eigenvalues of $\boldsymbol{\Sigma}^{-1}$ in increasing order are $1/\lambda_1 \leq \cdots \leq 1/\lambda_p$, so von Neumann's lower bound gives

$$\log|\boldsymbol{\Sigma}| + \mathrm{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_0) \geq \log(\lambda_1 \cdots \lambda_p) + \sum_{i=1}^p \frac{d_i}{\lambda_i} = \sum_{i=1}^p \Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big),$$

with equality iff $\mathbf{U} = \mathbf{V}$. Let $a_i = d_i/\lambda_i$. Then

$$\sum_{i=1}^p \Big(\log\lambda_i + \frac{d_i}{\lambda_i}\Big) = \sum_{i=1}^p (\log d_i + a_i - \log a_i) = \log|\mathbf{S}_0| + \sum_{i=1}^p (a_i - \log a_i) \geq \log|\mathbf{S}_0| + p,$$

since $x - \log x \geq 1$. The bound is attained iff (1) $\mathbf{U} = \mathbf{V}$ and (2) $a_i = 1$ for all i, i.e. $\boldsymbol{\Lambda} = \mathbf{D}$: $\hat{\boldsymbol{\Sigma}}_{\mathrm{MLE}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top = \mathbf{V}\mathbf{D}\mathbf{V}^\top = \mathbf{S}_0$. ∎

Second proof: matrix calculus. Reparametrise by the precision matrix $\mathbf{K} = \boldsymbol{\Sigma}^{-1}$ ($\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ is a bijection of the PD cone, so by the change-of-variables part of Theorem 2.22 the optimizers correspond). $g(\mathbf{K}) = -\log|\mathbf{K}| + \operatorname{tr}(\mathbf{K}\mathbf{S}_0)$ is defined on the open convex set $\{\mathbf{K} \succ 0\}$, and for symmetric $\mathbf{H} \neq \mathbf{0}$

$$D^2 g(\mathbf{K})[\mathbf{H}] = \operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}\mathbf{K}^{-1}\mathbf{H}) = \|\mathbf{K}^{-1/2}\mathbf{H}\mathbf{K}^{-1/2}\|_F^2 > 0,$$

so g is strictly convex (Theorem 2.21). The first-order condition

$$Dg(\mathbf{K})[\mathbf{H}] = -\operatorname{tr}(\mathbf{K}^{-1}\mathbf{H}) + \operatorname{tr}(\mathbf{S}_0\mathbf{H}) = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{K}^{-1})\mathbf{H}\} = 0 \quad \forall \mathbf{H}$$

and Theorem 2.20 give $\mathbf{K}^{-1} = \mathbf{S}_0$, the unique minimizer. By profiling (Theorem 2.22), $(\bar{\mathbf{x}}, \mathbf{S}_0)$ is the joint minimizer. ■

Remark 4.8: invariance and the maximized likelihood

A function of the MLE is the MLE of the function: e.g. the eigenvalues of $\mathbf{S}_0$ estimate those of $\boldsymbol{\Sigma}$, and r_{ij} is the MLE of ρ_{ij}. The maximum is $L(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$, which depends on the data only through the generalized variance $|\mathbf{S}_0|$; this is used for likelihood ratio tests (Ch. 5).

5 Large-Sample Behavior of $\bar{\mathbf{X}}$ and $\mathbf{S}$

Motivation. Without normality, the CLT still makes $\bar{\mathbf{X}}$ approximately normal and $\mathbf{S}$ consistent, so normal-theory procedures remain valid for large samples.

Theorem 4.9: law of large numbers

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_j) = \boldsymbol{\mu}$ and finite $\mathrm{Cov}(\mathbf{X}_j) = \boldsymbol{\Sigma}$. Then, as $n \to \infty$,

$$\text{(a) } \bar{\mathbf{X}}_n \xrightarrow{p} \boldsymbol{\mu}, \qquad \text{(b) } \mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}, \qquad \text{(c) } \mathbf{S}_0 \xrightarrow{p} \boldsymbol{\Sigma}.$$

Moreover $\bar{\mathbf{X}}_n$ and $\mathbf{S}$ are unbiased.

Proof. (a) By Markov's inequality,

$$P(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\| > \epsilon) \le \frac{1}{\epsilon^2}\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2 = \frac{1}{\epsilon^2}\mathrm{E}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \frac{1}{\epsilon^2}\mathrm{E}[\operatorname{tr}\{(\bar{\mathbf{X}}_n - \boldsymbol{\mu})(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\}]$$
$$= \frac{1}{\epsilon^2}\operatorname{tr}(\mathrm{Var}(\bar{\mathbf{X}}_n)) = \frac{\operatorname{tr}(\boldsymbol{\Sigma})}{n\epsilon^2} \xrightarrow[n\to\infty]{} 0.$$

(b) Adding and subtracting $\boldsymbol{\mu}$ inside each factor and using $\sum_i(\mathbf{X}_i - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$,

$$(n-1)\mathbf{S} = \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}}_n)(\mathbf{X}_i - \bar{\mathbf{X}}_n)^\top = \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top.$$

Hence

$$\frac{n-1}{n}\mathbf{S} = \frac{1}{n}\sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})^\top - (\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \xrightarrow{p} \boldsymbol{\Sigma} - \mathbf{0} = \boldsymbol{\Sigma}.$$

The first term is an average of iid matrices with mean $\mathrm{E}(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top = \boldsymbol{\Sigma}$, so it converges to $\boldsymbol{\Sigma}$ by the weak LLN applied entrywise (matrix convergence is entrywise, Theorem 2.24). The second term tends to $\mathbf{0}$ by (a) and the CMT ($\mathbf{x} \mapsto \mathbf{x}\mathbf{x}^\top$ is continuous). Since $n/(n-1) \to 1$, $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$.

(c) $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, which was shown to converge to $\boldsymbol{\Sigma}$ in (b).

Unbiasedness. $\mathrm{E}\bar{\mathbf{X}}_n = \boldsymbol{\mu}$, and taking expectations in the identity of (b), $\mathrm{E}\{(n-1)\mathbf{S}\} = n\boldsymbol{\Sigma} - n\mathrm{Var}(\bar{\mathbf{X}}_n) = (n-1)\boldsymbol{\Sigma}$. Under normality, $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ and Theorem 4.2(iv) also gives the rate $\mathrm{E}\|\mathbf{S} - \boldsymbol{\Sigma}\|_F^2 = \sum_{ij}(\Sigma_{ij}^2 + \Sigma_{ii}\Sigma_{jj})/(n-1) \to 0$. ■

Remark 4.10: rate of the estimation error

$\mathrm{E}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2 = \mathrm{tr}\boldsymbol{\Sigma}/n$, so $E_{\bar{u}} = \|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2$ decays at rate $n^{-1/2}$; so does $\|\mathbf{S} - \boldsymbol{\Sigma}\|_F$ under normality. On a log–log plot of $E_{\bar{u}}, E_S, E_R$ against n the slope is about $-1/2$.

Theorem 4.11: multivariate central limit theorem

Let $\mathbf{X}_1, \mathbf{X}_2, \cdots \in \mathbb{R}^p$ be iid with $\mathrm{E}(\mathbf{X}_i) = \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{X}_i) = \boldsymbol{\Sigma}$. Then

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\mathbf{X}_i - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \text{equivalently} \qquad \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Proof. By Cramér–Wold it suffices to show $\mathbf{t}^\top\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow \mathbf{t}^\top\mathbf{Y}$, $\mathbf{Y} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$, for every $\mathbf{t}$. $W_i = \mathbf{t}^\top(\mathbf{X}_i - \boldsymbol{\mu})$ are iid scalars with mean 0 and variance $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t}$, so the univariate CLT gives $\sqrt{n}\bar{W}_n \Rightarrow N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$, the law of $\mathbf{t}^\top\mathbf{Y}$ (if $\mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t} = 0$, $W_i = 0$ a.s. and the claim is trivial). ■

Theorem 4.12: large-sample chi-square

Under the same assumptions with $\boldsymbol{\Sigma} \succ 0$, $T_n := n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \Rightarrow \chi_p^2$. (Where $\mathbf{S}$ is singular, read $\mathbf{S}^{-1}$ as $r(\mathbf{S})$ defined in the proof.)

Proof. Let $\mathbf{Z}_n := \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ and $\mathbf{Z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.

(i) $\mathbf{Z}_n \Rightarrow \mathbf{Z}$. For every $\mathbf{t} \in \mathbb{R}^p$, $\mathbf{t}^\top\mathbf{Z}_n = \sqrt{n}(\mathbf{t}^\top\bar{\mathbf{X}}_n - \mathbf{t}^\top\boldsymbol{\mu}) \Rightarrow N(0, \mathbf{t}^\top\boldsymbol{\Sigma}\mathbf{t})$ by the univariate CLT, so $\mathbf{Z}_n \Rightarrow \mathbf{Z}$ by Cramér–Wold (this is Theorem 4.11).

(ii) Handling a possibly singular S. Let $A_n := \{\mathbf{S} \text{ is invertible}\}$. Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$ (Theorem 4.9(b)) and $\mathbf{A} \mapsto |\mathbf{A}|$ is continuous, $|\mathbf{S}| \xrightarrow{p} |\boldsymbol{\Sigma}| > 0$, hence

$$P(A_n) \geq P(|\mathbf{S}| \geq \tfrac{1}{2}|\boldsymbol{\Sigma}|) \longrightarrow 1.$$

Define $r(\mathbf{A}) = \mathbf{A}^{-1}$ if $|\mathbf{A}| \neq 0$ and $r(\mathbf{A}) = \mathbf{I}_p$ otherwise. By the definition of r,

$$T_n = n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n} + n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top(\bar{\mathbf{X}} - \boldsymbol{\mu})\, I_{A_n^c} = \mathbf{Z}_n^\top r(\mathbf{S})\mathbf{Z}_n,$$

and T_n coincides with the statistic of the theorem on A_n, an event of probability tending to one.

(iii) $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$. Since $\mathbf{S} \xrightarrow{p} \boldsymbol{\Sigma}$, by the CMT it suffices that r is continuous at the point $\boldsymbol{\Sigma}$. Since $|\boldsymbol{\Sigma}| > 0$ and $\{\mathbf{A} : |\mathbf{A}| \neq 0\}$ is open, $r(\mathbf{A}) = \mathbf{A}^{-1} = \mathrm{adj}(\mathbf{A})/|\mathbf{A}|$ on a neighborhood of $\boldsymbol{\Sigma}$ (Section 1 of Ch. 2), whose entries are polynomials divided by a nonvanishing polynomial; so r is continuous at $\boldsymbol{\Sigma}$ and $r(\mathbf{S}) \xrightarrow{p} \boldsymbol{\Sigma}^{-1}$.

(iv) Conclusion. Let $Q(\mathbf{z}, \mathbf{A}) = \mathbf{z}^\top\mathbf{A}\mathbf{z}$, so $T_n = Q(\mathbf{Z}_n, r(\mathbf{S}))$. By Slutsky, $(\mathbf{Z}_n, r(\mathbf{S})) \Rightarrow (\mathbf{Z}, \boldsymbol{\Sigma}^{-1})$, and Q is continuous, so by the CMT

$$T_n = Q(\mathbf{Z}_n, r(\mathbf{S})) \Rightarrow Q(\mathbf{Z}, \boldsymbol{\Sigma}^{-1}) = \mathbf{Z}^\top\boldsymbol{\Sigma}^{-1}\mathbf{Z} \sim \chi_p^2$$

(Example 3.22). ■

This is the basis of large-sample tests and confidence regions for the mean without normality (Theorems 5.16 and 5.17).

Theorem 4.13: asymptotic normality of $h(\bar{\mathbf{X}}_n)$ (delta method)

For $h : \mathbb{R}^p \to \mathbb{R}$ differentiable at $\boldsymbol{\mu}$ with gradient $\nabla h(\boldsymbol{\mu})$,

$$\sqrt{n}\{h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})\} \Rightarrow N_p(0,\ (\nabla h(\boldsymbol{\mu}))^\top \boldsymbol{\Sigma} (\nabla h(\boldsymbol{\mu}))).$$

Proof. Taylor: $h(\bar{\mathbf{X}}_n) = h(\boldsymbol{\mu}) + (\nabla h(\boldsymbol{\mu}))^\top (\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|_2^2)$. Multiply by $\sqrt{n}$: $\sqrt{n}(h(\bar{\mathbf{X}}_n) - h(\boldsymbol{\mu})) = (\nabla h(\boldsymbol{\mu}))^\top \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) + O_p(\sqrt{n}\|\bar{\mathbf{X}}_n - \boldsymbol{\mu}\|^2)$, and the remainder is $o_p(1)$. Slutsky gives $\Rightarrow (\nabla h(\boldsymbol{\mu}))^\top N_p(\mathbf{0}, \boldsymbol{\Sigma})$. ■

6 Summary: Distributional Facts Used Later

Fact	Source
$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$	Theorem 3.7
$(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$	Theorem 4.3
$(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_p$	Theorem 3.21, Example 3.22
$\mathbf{a}^\top \boldsymbol{\Sigma}^{-1} \mathbf{a} / \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a} \sim \chi^2_{m-p+1}$, also for independent random $\mathbf{y}$	Theorem 4.5
$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ (no normality)	Theorem 4.11 + Slutsky

Further topics: assessing normality (Q–Q plots; the χ^2 plot of $d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$ against χ^2_p quantiles), detecting outliers (large d_i^2), transformations to near normality (Box–Cox).

Chapter 5. Inference about a Mean Vector

Hotelling's T^2, likelihood ratio tests, confidence regions and simultaneous intervals, large-sample inference

1 Starting Point: the Univariate t and Known $\boldsymbol{\Sigma}$

In one dimension $H_0 : \mu = \mu_0$ is tested with $t = \sqrt{n}(\bar{X} - \mu_0)/s$, and

$$t_{n-1}^2 = \sqrt{n}(\bar{X} - \mu_0)\,(s^2)^{-1}\sqrt{n}(\bar{X} - \mu_0) = \begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}\begin{pmatrix}\text{(scaled) chi-square}\\ \text{r.v./d.f.}\end{pmatrix}^{-1}\begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}.$$

If $\boldsymbol{\Sigma}$ is known, $\sqrt{n}\boldsymbol{\Sigma}^{-1/2}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so

$$n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim \chi_p^2, \qquad C_\alpha = \{\boldsymbol{\mu} \in \mathbb{R}^p : (\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top (\tfrac{1}{n}\boldsymbol{\Sigma})^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \le \chi_{p,\alpha}^2\}.$$

Replacing $\boldsymbol{\Sigma}$ by $\mathbf{S}$ gives Hotelling's T^2.

2 Hotelling's T^2 Statistic

Motivation. Testing $\boldsymbol{\mu} = \boldsymbol{\mu}_0$ with p variables by p separate t-tests ignores correlation and inflates error. Hotelling's T^2 is the multivariate t-statistic: it combines all coordinates using $\mathbf{S}^{-1}$ and has an exact F distribution.

Definition 5.1: Hotelling's T^2

Suppose $\mathbf{X}$ and $\mathbf{S}$ are independent with $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$. Then

$$T_p^2(m) = (\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\mathbf{X} - \boldsymbol{\mu})$$

is Hotelling's T^2 statistic. It plays the role of Student's t when $\boldsymbol{\Sigma}$ is *unknown*:

$$T_p^2(n-1) = \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}^\top \begin{pmatrix}\text{Wishart random}\\ \text{matrix/d.f.}\end{pmatrix}^{-1} \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}$$

$$= N_p(\mathbf{0}, \boldsymbol{\Sigma})^\top \left[\tfrac{1}{n-1} W_p(n-1, \boldsymbol{\Sigma})\right]^{-1} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Theorem 5.2: T^2 and the F distribution

If $m > p - 1$, then $\dfrac{m - p + 1}{mp} T_p^2(m) \sim F_{p, m-p+1}$.

Proof. Multiplying and dividing by $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})$,

$$\frac{m-p+1}{mp} T_p^2(m) = \frac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})/p}{\dfrac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})}{(\mathbf{X} - \boldsymbol{\mu})^\top (m\mathbf{S})^{-1}(\mathbf{X} - \boldsymbol{\mu})} \Big/ (m - p + 1)} = \frac{Y/p}{W/(m-p+1)},$$

where $Y = (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2$ (Example 3.22) and $W = \dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top (m\mathbf{S})^{-1}\mathbf{a}}$ with the random vector $\mathbf{a} = \mathbf{X} - \boldsymbol{\mu}$ independent of $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$; by Theorem 4.5(iii) $W \sim \chi_{m-p+1}^2$ and is independent of $\mathbf{a}$, hence of Y. By definition the ratio is $F_{p, m-p+1}$. ■

Theorem 5.3: one-sample T^2

If $\mathbf{X}_1,\dots,\mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$, $\bar{\mathbf{X}}=\frac{1}{n}\sum\mathbf{X}_j$, $\mathbf{S}=\frac{1}{n-1}\sum(\mathbf{X}_j-\bar{\mathbf{X}})(\mathbf{X}_j-\bar{\mathbf{X}})^\top$, then $\sqrt{n}(\bar{\mathbf{X}}-\boldsymbol{\mu})\sim N_p(\mathbf{0},\boldsymbol{\Sigma})$ and $(n-1)\mathbf{S}\sim W_p(n-1,\boldsymbol{\Sigma})$ independently, so

$$n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu})\sim T^2_p(n-1),\qquad n(\bar{\mathbf{X}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{X}}-\boldsymbol{\mu})\sim\frac{(n-1)p}{n-p}F_{p,n-p}.$$

($m=n-1$ gives $\frac{m-p+1}{mp}=\frac{n-p}{(n-1)p}$.) Hence $H_0:\boldsymbol{\mu}=\boldsymbol{\mu}_0$ is rejected at level α if

$$T^2=n(\bar{\mathbf{x}}-\boldsymbol{\mu}_0)^\top\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}_0)>\frac{(n-1)p}{n-p}F_{p,n-p}(\alpha),$$

with p-value $P(F_{p,n-p}\geq\frac{n-p}{(n-1)p}T^2_{\text{obs}})$.

Theorem 5.4: invariance of T^2 under linear transformations

The T^2 statistic is unchanged by changes of units of the form $\mathbf{Y}_{(p\times1)}=\mathbf{C}_{(p\times p)}\mathbf{X}_{(p\times1)}+\mathbf{d}_{(p\times1)}$ with $\mathbf{C}$ non-singular: $\bar{\mathbf{y}}=\mathbf{C}\bar{\mathbf{x}}+\mathbf{d}$, $\mathbf{S}_y=\mathbf{C}\mathbf{S}\mathbf{C}^\top$, $\boldsymbol{\mu}_Y=\mathbf{C}\boldsymbol{\mu}_X+\mathbf{d}$, and

$$n(\bar{\mathbf{y}}-\boldsymbol{\mu}_Y)^\top\mathbf{S}_y^{-1}(\bar{\mathbf{y}}-\boldsymbol{\mu}_Y)=n(\bar{\mathbf{x}}-\boldsymbol{\mu}_x)^\top\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}_x)\sim T^2_p(n-1).$$

Proof. $n(\mathbf{C}(\bar{\mathbf{x}}-\boldsymbol{\mu}_x))^\top(\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1}\mathbf{C}(\bar{\mathbf{x}}-\boldsymbol{\mu}_x)=n(\bar{\mathbf{x}}-\boldsymbol{\mu}_x)^\top\mathbf{C}^\top\mathbf{C}^{-\top}\mathbf{S}^{-1}\mathbf{C}^{-1}\mathbf{C}(\bar{\mathbf{x}}-\boldsymbol{\mu}_x)$. ■

Example 5.5: testing a mean vector with T^2 (sweat data)

Perspiration from $n=20$ healthy females: X_1 sweat rate, X_2 sodium, X_3 potassium. Test $H_0:\boldsymbol{\mu}^\top=[4,50,10]$ at $\alpha=0.10$.

$$\bar{\mathbf{x}}=\begin{pmatrix}4.640\\45.400\\9.965\end{pmatrix},\quad \mathbf{S}=\begin{pmatrix}2.879&10.010&-1.810\\10.010&199.788&-5.640\\-1.810&-5.640&3.628\end{pmatrix},\quad \mathbf{S}^{-1}=\begin{pmatrix}.586&-.022&.258\\-.022&.006&-.002\\.258&-.002&.402\end{pmatrix}.$$

$\bar{\mathbf{x}}-\boldsymbol{\mu}_0=(.640,-4.600,-.035)^\top$ and $\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}_0)=(.467,-.042,.160)^\top$, so $T^2=20(.640\cdot.467+4.600\cdot.042-.035\cdot.160)=9.74$. Critical value $\frac{(n-1)p}{n-p}F_{p,n-p}(.10)=\frac{19(3)}{17}F_{3,17}(.10)=3.353(2.44)=8.18$. Since $9.74>8.18$, reject H_0 at the 10% level.

3 Likelihood Ratio Tests

3.1 $\boldsymbol{\Sigma}$ unknown

Under $H_0:\boldsymbol{\mu}=\boldsymbol{\mu}_0$ the likelihood is

$$L(\boldsymbol{\mu}_0,\boldsymbol{\Sigma})=\frac{1}{(2\pi)^{np/2}|\boldsymbol{\Sigma}|^{n/2}}\exp\Big(-\frac{1}{2}\sum_{j=1}^{n}(\mathbf{x}_j-\boldsymbol{\mu}_0)^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}_j-\boldsymbol{\mu}_0)\Big),$$

and the exponent equals $-\frac{1}{2}\text{tr}[\boldsymbol{\Sigma}^{-1}\sum_j(\mathbf{x}_j-\boldsymbol{\mu}_0)(\mathbf{x}_j-\boldsymbol{\mu}_0)^\top]$. Maximizing over $\boldsymbol{\Sigma}$ as in the proof of Theorem 4.7 (with $\sum_j(\mathbf{x}_j-\boldsymbol{\mu}_0)(\mathbf{x}_j-\boldsymbol{\mu}_0)^\top$ in place of $\sum_j(\mathbf{x}_j-\bar{\mathbf{x}})(\mathbf{x}_j-\bar{\mathbf{x}})^\top$),

$$\max_{\boldsymbol{\Sigma}}L(\boldsymbol{\mu}_0,\boldsymbol{\Sigma})=\frac{1}{(2\pi)^{np/2}|\hat{\boldsymbol{\Sigma}}_0|^{n/2}}e^{-np/2},\qquad\hat{\boldsymbol{\Sigma}}_0=\frac{1}{n}\sum_{j=1}^{n}(\mathbf{x}_j-\boldsymbol{\mu}_0)(\mathbf{x}_j-\boldsymbol{\mu}_0)^\top.$$

The unrestricted maximum has the same form with $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$, so the *likelihood ratio statistic* is

$$\Lambda = \frac{\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})}{\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L(\boldsymbol{\mu}, \boldsymbol{\Sigma})} = \Big(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}\Big)^{n/2}, \qquad \Lambda^{2/n} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} = \frac{|\sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top|}{|\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top|}.$$

$\Lambda^{2/n}$ is *Wilks' lambda*. Reject H_0 if $\Lambda < c_\alpha$, the lower 100αth percentile of its distribution; fortunately the distribution of $\Lambda^{2/n}$ is available.

Theorem 5.6: likelihood ratio test and T^2

For a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the T^2 test is equivalent to the likelihood ratio test of $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ because

$$\Lambda^{2/n} = \Big(1 + \frac{T^2}{n-1}\Big)^{-1}.$$

H_0 is rejected for small $\Lambda^{2/n}$, equivalently large T^2.

First proof: block determinant. Let $\mathbf{Q} = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$, $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$, and consider the $(p+1) \times (p+1)$ matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{Q} & \sqrt{n}\,\boldsymbol{\delta} \\ \sqrt{n}\,\boldsymbol{\delta}^\top & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}.$$

From $|\mathbf{A}| = |\mathbf{A}_{22}||\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}||\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}|$,

$$(-1)|\mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{Q}|(-1 - n\boldsymbol{\delta}^\top\mathbf{Q}^{-1}\boldsymbol{\delta}) = -|\mathbf{Q}|\Big(1 + \frac{T^2}{n-1}\Big).$$

Also $\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}} + \bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\cdot)^\top = \mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top$. Hence $|n\hat{\boldsymbol{\Sigma}}_0| = |n\hat{\boldsymbol{\Sigma}}|(1 + \frac{T^2}{n-1})$ and $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0| = (1 + \frac{T^2}{n-1})^{-1}$. ■

Second proof: determinant lemma. With $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$ and $\mathbf{S}_0 = \hat{\boldsymbol{\Sigma}}$, $\hat{\boldsymbol{\Sigma}}_0 = \mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ and $|\mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{S}_0^{1/2}||\mathbf{I}_p + \mathbf{S}_0^{-1/2}\boldsymbol{\delta}\boldsymbol{\delta}^\top\mathbf{S}_0^{-1/2}||\mathbf{S}_0^{1/2}| = |\mathbf{S}_0|(1 + \boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta})$ using $|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u}$. Since $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, $\boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta} = \frac{n}{n-1}\boldsymbol{\delta}^\top\mathbf{S}^{-1}\boldsymbol{\delta} = \frac{T^2}{n-1} =: \frac{T_0^2}{n-1}$. ■

Remark 5.7: approximation of $W = -2\log\Lambda$

$W = -2\log(L_0/L_1) = n\log(1 + \frac{T^2}{n-1})$. Under H_0, $T^2 = O_p(1)$, so $W \approx \frac{n}{n-1}T^2 \approx T^2$, and for large n, $\frac{np}{n-p}F_{p,n-p} \approx \chi_p^2$: $W \dot{\sim} \chi_p^2$, in agreement with general likelihood ratio theory. Exact test: reject if $T_0^2 > \frac{(n-1)p}{n-p}F_{p,n-p,\alpha}$; approximate test: reject if $W > \chi_{p,\alpha}^2$.

3.2 $\boldsymbol{\Sigma}$ known

With $L_0 = L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$ and $L_1 = L(\bar{\mathbf{x}}, \boldsymbol{\Sigma})$, $W = -2\log L_0 - (-2\log L_1) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$, which is exactly χ_p^2 under H_0.

3.3 Testing a covariance matrix: $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$, $\boldsymbol{\mu}$ unknown

Theorem 5.8: likelihood ratio test for $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$

Let $\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu}$ unknown, and $\boldsymbol{\Sigma}_0 \succ 0$ given. For $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$ versus $H_1 : \boldsymbol{\Sigma} \neq \boldsymbol{\Sigma}_0$,

$$W = -2 \log \Lambda = -n \log |\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| - np + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) \;\dot{\sim}\; \chi^2_\nu, \qquad \nu = \frac{p(p+1)}{2},$$

where $\mathbf{S}_0 = \frac{1}{n} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. Equivalently, with $l_1, \dots, l_p$ the eigenvalues of $\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0$,

$$W = n \sum_{i=1}^{p} (l_i - \log l_i - 1) \;\geq 0.$$

Reject H_0 if $W > \chi^2_\nu(\alpha)$.

Proof [Supplement]. Step 1 (the statistic). By Theorem 4.7, $-2 \log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = n \log |\boldsymbol{\Sigma}| + n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}) + n \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S}_0) + c$. Under H_1 the maximum is at $(\bar{\mathbf{x}}, \mathbf{S}_0)$: $-2 \log L_1 = n \log |\mathbf{S}_0| + np + c$. Under H_0 only $\boldsymbol{\mu}$ is free and its maximizer is $\bar{\mathbf{x}}$ whatever $\boldsymbol{\Sigma}_0$ is: $-2 \log L_0 = n \log |\boldsymbol{\Sigma}_0| + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) + c$. Hence

$$\begin{aligned} W = -2 \log L_0 + 2 \log L_1 &= n \log |\boldsymbol{\Sigma}_0| - n \log |\mathbf{S}_0| + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) - np \\ &= -n \log |\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| - np + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0). \end{aligned}$$

The eigenvalues l_i of $\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0$ equal those of the PD matrix $\boldsymbol{\Sigma}_0^{-1/2} \mathbf{S}_0 \boldsymbol{\Sigma}_0^{-1/2}$, so $l_i > 0$, $|\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| = \prod l_i$, $\operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) = \sum l_i$, giving $W = n \sum (l_i - \log l_i - 1) \geq 0$ by Theorem 2.13, with $W = 0$ iff $\mathbf{S}_0 = \boldsymbol{\Sigma}_0$.

Step 2 (reduction to $\boldsymbol{\Sigma}_0 = \mathbf{I}$). Put $\mathbf{Y}_i = \boldsymbol{\Sigma}_0^{-1/2} \mathbf{X}_i$. Then $\mathbf{S}_0^{(Y)} = \boldsymbol{\Sigma}_0^{-1/2} \mathbf{S}_0 \boldsymbol{\Sigma}_0^{-1/2}$ has the same eigenvalues l_i, so W is unchanged, and under H_0, $\mathbf{Y}_i \overset{\text{iid}}{\sim} N_p(\boldsymbol{\Sigma}_0^{-1/2} \boldsymbol{\mu}, \mathbf{I}_p)$. So assume $\boldsymbol{\Sigma}_0 = \mathbf{I}$ and write $\mathbf{S}_0$ for the MLE.

Step 3 (quadratic approximation). Under H_0, $\mathbf{S}_0 \xrightarrow{p} \mathbf{I}$, so all $l_i \xrightarrow{p} 1$. Taylor expansion of $g(l) = l - \log l - 1$ at $l = 1$: $g(1) = g'(1) = 0$, $g''(1) = 1$, so $g(l) = \frac{1}{2}(l-1)^2 + O(|l-1|^3)$ and

$$W = \frac{n}{2} \sum_i (l_i - 1)^2 + n \, O_p\Big(\max_i |l_i - 1|^3 \Big) = \frac{n}{2} \|\mathbf{S}_0 - \mathbf{I}\|_F^2 + o_p(1),$$

using $\sum_i (l_i - 1)^2 = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{I})^2\} = \|\mathbf{S}_0 - \mathbf{I}\|_F^2$ and $\sqrt{n}(\mathbf{S}_0 - \mathbf{I}) = O_p(1)$ (Step 4), so that $n \max |l_i - 1|^3 = O_p(n^{-1/2})$.

Step 4 (limit of $\sqrt{n}(\mathbf{S}_0 - \mathbf{I})$). Write $\mathbf{S}_0 = \frac{1}{n} \sum_i \mathbf{Z}_i \mathbf{Z}_i^\top - (\bar{\mathbf{Z}})(\bar{\mathbf{Z}})^\top$ with $\mathbf{Z}_i = \mathbf{Y}_i - \mathrm{E}\mathbf{Y}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{I})$; the second term is $O_p(1/n)$. For the vector of distinct entries $\{Z_j Z_k\}_{j \leq k}$, $\mathrm{E}(Z_j Z_k) = \delta_{jk}$ and

$$\operatorname{Cov}(Z_j Z_k, Z_l Z_m) = \mathrm{E}(Z_j Z_k Z_l Z_m) - \delta_{jk} \delta_{lm} = \delta_{jl} \delta_{km} + \delta_{jm} \delta_{kl},$$

so $\operatorname{Var}(Z_j^2) = 2$, $\operatorname{Var}(Z_j Z_k) = 1$ $(j < k)$, and different pairs $\{j, k\} \neq \{l, m\}$ are uncorrelated (consistent with Theorem 4.2(iv): $\Sigma_{jk}^2 + \Sigma_{jj} \Sigma_{kk}$). By the multivariate CLT,

$$\sqrt{n}(s_{jj} - 1) \Rightarrow N(0, 2), \qquad \sqrt{n}\, s_{jk} \Rightarrow N(0, 1) \;\; (j < k),$$

jointly, with a diagonal limiting covariance, i.e. all $p(p+1)/2$ limits are independent.

Step 5 (chi-square limit). By the CMT,

$$\frac{n}{2} \|\mathbf{S}_0 - \mathbf{I}\|_F^2 = \sum_{j=1}^{p} \frac{\{\sqrt{n}(s_{jj} - 1)\}^2}{2} + \sum_{j<k} \{\sqrt{n}\, s_{jk}\}^2 \Rightarrow \sum_{j=1}^{p} \frac{(\sqrt{2} Z_{jj})^2}{2} + \sum_{j<k} Z_{jk}^2 = \chi^2_{p + p(p-1)/2} = \chi^2_{p(p+1)/2},$$

with independent $Z_{jk} \sim N(0,1)$. By Slutsky, $W \Rightarrow \chi^2_{p(p+1)/2}$. The degrees of freedom $p(p+1)/2$ are the number of free parameters in a symmetric $\boldsymbol{\Sigma}$ fixed by H_0, as general likelihood ratio theory predicts. ■

Remark 5.9: special cases

- $\boldsymbol{\Sigma}_0 = \sigma_0^2 \mathbf{I}$: $W = n\{\mathrm{tr}(\mathbf{S}_0)/\sigma_0^2 - \log|\mathbf{S}_0/\sigma_0^2| - p\}$.
- $p = 1$: $W = n\{s_0^2/\sigma_0^2 - \log(s_0^2/\sigma_0^2) - 1\}$, an asymptotic version of the χ^2 test for a variance; the exact test uses $ns_0^2/\sigma_0^2 \sim \chi^2_{n-1}$.
- Bartlett-type corrections replace n by $n-1$ (using $\mathbf{S}$ instead of $\mathbf{S}_0$) and an additional factor to improve the χ^2 approximation.

3.4 Summary: likelihood ratio tests under normality

Hypothesis	Statistic and distribution
(i) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, $\boldsymbol{\Sigma}$ known	$W = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) \sim \chi^2_p$ (exact)
(ii) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, $\boldsymbol{\Sigma}$ unknown	$W = n\log(1 + T^2/(n-1))$; exact: $T^2 \sim \frac{(n-1)p}{n-p} F_{p,n-p}$
(iii) $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$, $\boldsymbol{\mu}$ unknown	$W = -n\log\lvert\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0\rvert - np + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) \dot{\sim} \chi^2_{p(p+1)/2}$
(iv) $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, common $\boldsymbol{\Sigma}$ unknown	$W = n\log\left(1 + \frac{T^2}{n_1+n_2-2}\right)$; exact: $\frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2 \sim F_{p,n_1+n_2-p-1}$

Theorem 5.10: two-sample likelihood ratio test

Let $\mathbf{X}_{11}, \ldots, \mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $\mathbf{X}_{21}, \ldots, \mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$ be independent, $n = n_1 + n_2$. For $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$ with $\boldsymbol{\Sigma}$ unknown,

$$\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \sim N_p\Big(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, \frac{n_1+n_2}{n_1 n_2}\boldsymbol{\Sigma}\Big), \qquad (n_1+n_2-2)\mathbf{S}_p \sim W_p(n_1+n_2-2, \boldsymbol{\Sigma}),$$

$$T^2(n_1+n_2-2) = \frac{n_1 n_2}{n_1+n_2}(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2)^\top \mathbf{S}_p^{-1}(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2),$$

$$\frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2(n_1+n_2-2) \sim F_{p,n_1+n_2-p-1},$$

and the likelihood ratio satisfies $\Lambda^{2/n} = \left(1 + \frac{T^2}{n_1+n_2-2}\right)^{-1}$, so the LRT is the T^2 test.

Proof [Supplement]. The distributional facts are Theorem 6.5. Under H_1 the MLEs are $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2$ and $\hat{\boldsymbol{\Sigma}}_1 = \mathbf{W}/n$, $\mathbf{W} = (n_1+n_2-2)\mathbf{S}_p$; under H_0 they are the pooled mean $\bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}}_0 = (\mathbf{W}+\mathbf{B})/n$ with $\mathbf{B} = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top$, $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$ (the $g = 2$ case of Ch. 6). As in Theorem 5.6, $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}_1|/|\hat{\boldsymbol{\Sigma}}_0| = |\mathbf{W}|/|\mathbf{W}+\mathbf{B}|$, and the determinant lemma gives $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1 + \frac{n_1 n_2}{n}\mathbf{d}^\top \mathbf{W}^{-1}\mathbf{d}) = |\mathbf{W}|(1 + \frac{T^2}{n_1+n_2-2})$. ■

Note. The F constant is $\frac{n_1+n_2-p-1}{(n_1+n_2-2)p}$, from Theorem 5.2 with $m = n_1 + n_2 - 2$.

4 Confidence Regions

Motivation. Inverting the T^2 test gives a confidence ellipsoid for $\boldsymbol{\mu}$, whose axes are the eigenvectors of $\mathbf{S}$.

Theorem 5.11: $100(1-\alpha)\%$ confidence region for $\boldsymbol{\mu}$

$$R(\mathbf{X}) = \Big\{\boldsymbol{\mu} \in \mathbb{R}^p : \ \frac{n}{n-1}\frac{n-p}{p}(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top \mathbf{S}_0^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le F_{p,n-p,\alpha}\Big\}$$
$$= \Big\{\boldsymbol{\mu} : n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)\Big\}.$$

With $\mathbf{S} = \sum \lambda_i \mathbf{e}_i \mathbf{e}_i^\top$ it is an ellipsoid centered at $\bar{\mathbf{x}}$ with axes along $\mathbf{e}_i$ and half-lengths $\sqrt{\lambda_i}\sqrt{\frac{(n-1)p}{n(n-p)}F_{p,n-p}(\alpha)}$. Checking whether $\boldsymbol{\mu}_0 \in R$ is the T^2 test.

For $p \ge 3$ the ellipsoid cannot be drawn, so componentwise intervals are needed.

5 Simultaneous Confidence Intervals

5.1 A single linear combination

For fixed $\mathbf{a}$, $\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu}) \sim N_1(0, \frac{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}{n})$ and $(n-1)\mathbf{a}^\top\mathbf{S}\mathbf{a} \sim W_1(n-1, \mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}) \iff (n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}} \sim \chi^2_{n-1}$, independent, so

$$t(\mathbf{a}) = \frac{\sqrt{n}\,\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})/\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}}{\sqrt{(n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}/(n-1)}} = \frac{\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \sim t_{n-1}, \qquad C_\alpha = \Big\{\theta : \frac{|\mathbf{a}^\top\bar{\mathbf{x}}-\theta|}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \le t_{n-1,\alpha/2}\Big\}.$$

Using these intervals for several $\mathbf{a}$ at once lowers the overall confidence below $1-\alpha$.

5.2 T^2 simultaneous intervals

Theorem 5.12: simultaneous confidence intervals for $\mathbf{a}^\top\boldsymbol{\mu}$

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$. Then, *simultaneously for all* $\mathbf{a}$, the interval

$$\Big(\mathbf{a}^\top\bar{\mathbf{X}} - \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}}, \ \ \mathbf{a}^\top\bar{\mathbf{X}} + \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}}\Big)$$

contains $\mathbf{a}^\top\boldsymbol{\mu}$ with probability $1-\alpha$.

Proof. Simple algebra gives, for every $\mathbf{a}$,

$$\mathbf{a}^\top\bar{\mathbf{x}} - c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \le \mathbf{a}^\top\boldsymbol{\mu} \le \mathbf{a}^\top\bar{\mathbf{x}} + c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \iff t(\mathbf{a})^2 = \frac{n(\mathbf{a}^\top\bar{\mathbf{x}}-\mathbf{a}^\top\boldsymbol{\mu})^2}{\mathbf{a}^\top\mathbf{S}\mathbf{a}} \le c^2.$$

The key is $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$: with $\mathbf{b} = \mathbf{S}^{1/2}\mathbf{a}$ and $\mathbf{v} = \sqrt{n}\mathbf{S}^{-1/2}(\bar{\mathbf{x}}-\boldsymbol{\mu})$,

$$\max_{\mathbf{a}} t(\mathbf{a})^2 = \max_{\mathbf{b}} \frac{(\mathbf{b}^\top\mathbf{v})^2}{\mathbf{b}^\top\mathbf{b}} \le \mathbf{v}^\top\mathbf{v} = n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) = T^2(n-1)$$

by Cauchy–Schwarz, with equality at $\mathbf{a} \propto \mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu})$. By Theorem 5.2, choosing $c^2 = \frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)$ gives $P(\bigcap_{\mathbf{a}}\{t(\mathbf{a})^2 \le c^2\}) = P(\max_{\mathbf{a}} t(\mathbf{a})^2 \le c^2) = P(T^2 \le c^2) = 1-\alpha$. ∎

- These are called T^2-intervals, since the coverage is determined by the distribution of T^2. With $\mathbf{a} = \mathbf{e}_j$: $\bar{x}_j \pm \sqrt{\frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)}\sqrt{\frac{s_{jj}}{n}}$, holding simultaneously with confidence $1-\alpha$.

- With $\mathbf{a}^\top = [0,\dots,0,a_j,0,\dots,0,a_k,0,\dots]$, $a_j = 1$, $a_k = -1$: $\mathbf{a}^\top\mathbf{S}\mathbf{a} = s_{jj} - 2s_{jk} + s_{kk}$, giving intervals for $\mu_j - \mu_k$.
- The coefficient $1-\alpha$ holds for any choice of $\mathbf{a}$, so these intervals are ideal for "data snooping".
- The rectangle formed by the componentwise T^2 intervals is the shadow of the confidence ellipsoid on the coordinate axes.

Example 5.13: simultaneous intervals (microwave radiation data)

$n = 42$, $p = 2$, $\bar{\mathbf{x}} = (0.564, 0.603)^\top$, $s_{11} = 0.0144$, $s_{22} = 0.0146$, $F_{2,40}(.05) = 3.23$. The 95% simultaneous T^2 intervals are

$$0.564 \pm \sqrt{\tfrac{2(41)}{40}3.23}\sqrt{\tfrac{0.0144}{42}} = (0.516, 0.612), \qquad 0.603 \pm \sqrt{\tfrac{2(41)}{40}3.23}\sqrt{\tfrac{0.0146}{42}} = (0.555, 0.651).$$

5.3 The Bonferroni method

When attention is restricted to a small number m of pre-specified linear combinations, shorter (more precise) simultaneous intervals are possible.

Theorem 5.14: Bonferroni inequality

Let C_i be a confidence statement about $\mathbf{a}_i^\top\boldsymbol{\mu}$ with $P(C_i \text{ true}) = 1 - \alpha_i$. Then

$$P[\text{all } C_i \text{ true}] = 1 - P[\text{at least one } C_i \text{ false}] \geq 1 - \sum_{i=1}^{m} P(C_i \text{ false}) = 1 - (\alpha_1 + \cdots + \alpha_m).$$

With $\alpha_i = \alpha/m$, the individual intervals $\bar{x}_i \pm t_{n-1}(\frac{\alpha}{2m})\sqrt{s_{ii}/n}$, $i = 1,\dots,m$, hold simultaneously with confidence at least $1-\alpha$. (Equivalently, with $C_i(\alpha) = \{\mu_i : |\bar{x}_i - \mu_i|/(s_i/\sqrt{n}) < t_{n-1,\alpha/2}\}$; setting $\mu_i \in C_i(\alpha/m)$ gives $P(\mu_i \in C_i(\alpha/m),\ i = 1,\dots,m) \geq 1 - \sum_i P(\mu_i \notin C_i(\alpha/m)) = 1 - \alpha$.)

Remark 5.15: comparison of Bonferroni and T^2 interval lengths

Both have the form $\mathbf{a}^\top\bar{\mathbf{x}} \pm (\text{critical value})\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}$, so with $\alpha_i = \alpha/m$

$$\frac{\text{length of Bonferroni interval}}{\text{length of } T^2 \text{ interval}} = \frac{t_{n-1}(\alpha/2m)}{\sqrt{\frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)}},$$

which does not depend on $\bar{\mathbf{X}}$ or $\mathbf{S}$. For $1-\alpha = .95$, $\alpha_i = .05/m$, $m = p$:

n	$m=2$	$m=4$	$m=10$
15	.88	.69	.29
25	.90	.75	.48
50	.91	.78	.58
100	.91	.80	.62
∞	.91	.81	.66

For a small number of specified functions $\mathbf{a}^\top\boldsymbol{\mu}$ the Bonferroni intervals are shorter; being easy to apply, they are used often. In Example 5.13 the Bonferroni intervals are $(0.521, 0.607)$ and $(0.560, 0.646)$, inside the T^2 intervals.

6 Large-Sample Inference about a Mean Vector

Motivation. When normality is doubtful but n is large, T^2 is approximately χ^2_p, and the same procedures apply with χ^2 critical values.

Theorem 5.16: hypothesis testing for $\boldsymbol{\mu}$ with large $n-p$

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a population with mean $\boldsymbol{\mu}$ and positive definite covariance $\boldsymbol{\Sigma}$ (not necessarily normal). When $n-p$ is large, $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected in favor of $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, at level approximately α, if

$$n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \chi^2_p(\alpha).$$

The basis is $n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ (Ch. 4). The statistic is the same as in normal theory; only the critical values differ, since for n large relative to p, $\frac{(n-1)pF_{p,n-p}(\alpha)}{n-p} \approx \chi^2_p(\alpha)$.

Theorem 5.17: large-sample simultaneous intervals

If $n-p$ is large, $\mathbf{a}^\top \bar{\mathbf{X}} \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top \mathbf{S}\mathbf{a}/n}$ contains $\mathbf{a}^\top \boldsymbol{\mu}$ for every $\mathbf{a}$ with probability approximately $1-\alpha$. In particular $\bar{x}_j \pm \sqrt{\chi^2_p(\alpha)}\sqrt{s_{jj}/n}$ contains μ_j, $j = 1, \ldots, p$.

What counts as "large" is not easy to answer. Example: from the Golub data take two genes ($p = 2$) and the 11 AML and 27 ALL patients; the 95% and 99% elliptical regions and the simultaneous intervals (with $\mathbf{a}_1 = (1,0)^\top$, $\mathbf{a}_2 = (0,1)^\top$) show clearly different means for the two groups.

Further topic: multivariate quality control charts — the T^2 chart compares $T^2_j = (\mathbf{x}_j - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}})$ with the control limit $\chi^2_p(\alpha)$.

Chapter 6. Comparing Mean Vectors and MANOVA

Paired comparisons, contrasts, two populations, one- and two-way MANOVA, simultaneous intervals

1 Paired Comparisons and Repeated Measures

1.1 Paired comparisons

When two treatments are applied to the same unit and p responses are measured, analyze the differences $\mathbf{D}_j = \mathbf{X}_{1j} - \mathbf{X}_{2j}$, $j = 1, \dots, n$; unit effects cancel, increasing precision.

Theorem 6.1: paired T^2

Let $\mathbf{D}_1, \dots, \mathbf{D}_n$ be a random sample from $N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_d)$. Then

$$T^2 = n(\bar{\mathbf{D}} - \boldsymbol{\delta})^\top \mathbf{S}_d^{-1} (\bar{\mathbf{D}} - \boldsymbol{\delta})$$

is distributed as $\frac{(n-1)p}{n-p} F_{p,n-p}$, whatever the true $\boldsymbol{\delta}$ and $\boldsymbol{\Sigma}_d$. If n and $n-p$ are both large, T^2 is approximately χ^2_p regardless of the population of differences.

This is the one-sample T^2 applied to $\mathbf{D}_j$; note $(n-1)\mathbf{S}_d \sim W_p(n-1, \boldsymbol{\Sigma}_d)$. The test of $H_0 : \boldsymbol{\delta} = \mathbf{0}$, the confidence region, simultaneous intervals $\bar{d}_i \pm \sqrt{\frac{(n-1)p}{n-p} F_{p,n-p}(\alpha)} \sqrt{s^2_{d_i}/n}$ and Bonferroni intervals $\bar{d}_i \pm t_{n-1}(\alpha/2p)\sqrt{s^2_{d_i}/n}$ are as in Ch. 5.

1.2 Contrasts and repeated measures designs

With q repeated measurements $\mathbf{X}_j \in \mathbb{R}^q$ on each unit, "all treatment means are equal" is a hypothesis $\mathbf{C}\boldsymbol{\mu} = \mathbf{0}$, e.g. $\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 & \cdots \\ -1 & 0 & 1 & \cdots \\ & & \ddots & \end{pmatrix}$ ($(q-1) \times q$, rows are contrasts).

Theorem 6.2: test for contrasts

If $\mathbf{X}_j \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{C} \in \mathbb{R}^{q \times p}$ has rank q, then under $H_0 : \mathbf{C}\boldsymbol{\mu} = \mathbf{b}$

$$T_c^2 = n(\mathbf{C}\bar{\mathbf{x}} - \mathbf{b})^\top (\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1} (\mathbf{C}\bar{\mathbf{x}} - \mathbf{b}) \sim \frac{(n-1)q}{n-q} F_{q,n-q} = \frac{q(n-1)}{(n-1)-(q-1)} F_{q,n-q}.$$

Proof. (i) $\sqrt{n}(\mathbf{C}\bar{\mathbf{X}} - \mathbf{C}\boldsymbol{\mu}) \sim N_q(\mathbf{0}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$;
(ii) $(n-1)\mathbf{C}\mathbf{S}\mathbf{C}^\top \sim W_q(n-1, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\top)$ (Theorem 4.2(i)), independent of $\mathbf{C}\bar{\mathbf{X}}$ since $\bar{\mathbf{X}} \perp\!\!\!\perp \mathbf{S}$. Apply Theorem 5.2 in dimension q with $m = n-1$. ■

By invariance (Ch. 5), T_c^2 does not depend on which contrast matrix with the same row space is used.

2 Comparing Mean Vectors from Two Populations

Motivation. Two-group comparisons generalize the two-sample t-test. With a common covariance matrix, the pooled $\mathbf{S}$ gives an exact T^2 test; with unequal covariances, large-sample approximations

are used.

Remark 6.3: comparing two distributions versus two means

The general question is $H_0 : F_1 = F_2$ versus $H_1 : F_1 \neq F_2$ for the two population distributions. If both populations are normal with a common covariance, $F_\ell = N_p(\boldsymbol{\mu}_\ell, \boldsymbol{\Sigma})$, then $F_1 = F_2 \iff \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, so the problem reduces to $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$. Without these assumptions, equal means does not imply equal distributions.

Remark 6.4: the univariate case $p = 1$

With samples $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$, the pooled two-sample t statistic is

$$T = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1/2} \frac{\bar{X}_m - \bar{Y}_n}{S_p}, \qquad S_p^2 = \frac{\sum_{i=1}^m (X_i - \bar{X}_m)^2 + \sum_{i=1}^n (Y_i - \bar{Y}_n)^2}{m + n - 2},$$

and $T \sim t_{m+n-2}$ under H_0. For $p \geq 1$ the pooled covariance $\mathbf{S}_p = (s_{ij})$ has entries

$$s_{ij} = \frac{\sum_{k=1}^m (X_{ki} - \bar{X}_{\cdot i})(X_{kj} - \bar{X}_{\cdot j}) + \sum_{k=1}^n (Y_{ki} - \bar{Y}_{\cdot i})(Y_{kj} - \bar{Y}_{\cdot j})}{m + n - 2},$$

and the multivariate statistic is the squared norm generalizing T^2:

$$T^2 = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1} \left\| \mathbf{S}_p^{-1/2} (\bar{\mathbf{X}}_m - \bar{\mathbf{Y}}_n) \right\|_2^2 \sim \frac{(m + n - 2)p}{m + n - p - 1} F_{p, m+n-p-1}.$$

For $p = 1$ this is exactly the square of the t statistic, since $t_\nu^2 = F_{1,\nu}$. The p-value is $P(T^2 > T^2_{\text{obs}} \mid H_0) = P(F_{p,m+n-p-1} \geq F_{\text{obs}})$ with $F_{\text{obs}} = \frac{m+n-p-1}{(m+n-2)p} T^2_{\text{obs}}$. This follows from the general form $T^2 = \mathbf{z}^\top (\mathbf{W}/\nu)^{-1} \mathbf{z} \sim \frac{p\nu}{\nu-(p-1)} F_{p,\nu-(p-1)}$ for $\mathbf{z} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ independent of $\mathbf{W} \sim W_p(\nu, \boldsymbol{\Sigma})$ (Theorem 5.2 with $m = \nu$), here with $\nu = m + n - 2$.

2.1 Common covariance: the pooled T^2

$\mathbf{X}_{11}, \ldots, \mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and an independent sample $\mathbf{X}_{21}, \ldots, \mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$. Pooled covariance:

$$\mathbf{S}_{\text{pooled}} = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}.$$

Theorem 6.5: two-sample T^2

$$T^2 = (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))^\top \left[\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)\mathbf{S}_{\text{pooled}}\right]^{-1} (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2))$$
$$\sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1+n_2-p-1}.$$

Consequently $P(T^2 \leq c_\alpha^2) = 1 - \alpha$ with $c_\alpha^2 = \frac{(n_1+n_2-2)p}{n_1+n_2-p-1} F_{p,n_1+n_2-p-1}(\alpha)$.

Proof. (i) $\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \sim N_p(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, (\frac{1}{n_1} + \frac{1}{n_2})\boldsymbol{\Sigma})$, so $\mathbf{z} = (\frac{1}{n_1} + \frac{1}{n_2})^{-1/2}\{\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$.

(ii) $(n_i - 1)\mathbf{S}_i \sim W_p(n_i - 1, \boldsymbol{\Sigma})$ independently, so $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}} \sim W_p(n_1 + n_2 - 2, \boldsymbol{\Sigma})$ by Theorem 4.2(iii).

(iii) $\bar{\mathbf{X}}_i \perp\!\!\!\perp \mathbf{S}_i$ within each sample and the samples are independent, so $\mathbf{z} \perp\!\!\!\perp \mathbf{S}_{\text{pooled}}$. Hence $T^2 = \mathbf{z}^\top \mathbf{S}_{\text{pooled}}^{-1} \mathbf{z} = (\frac{1}{n_1} + \frac{1}{n_2})^{-1} \| \mathbf{S}_p^{-1/2} (\bar{\mathbf{x}} - \bar{\mathbf{y}} - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)) \|_2^2 \sim T_p^2(n_1 + n_2 - 2)$, and Theorem 5.2 with $m = n_1 + n_2 - 2$ gives the F form. ∎

The p-value is $P(F_{p,n_1+n_2-p-1} \geq F_{\text{obs}})$ with $F_{\text{obs}} = \frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2_{\text{obs}}$. For contrasts, $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{C}\boldsymbol{\mu}_2$ ($\mathbf{C}$: $q \times p$):

$$\left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right)^{-1} \|(\mathbf{C}\mathbf{S}_p\mathbf{C}^\top)^{-1/2}\mathbf{C}(\bar{\mathbf{x}} - \bar{\mathbf{y}})\|_2^2 \sim \frac{q(n_1+n_2-2)}{n_1+n_2-q-1} F_{q,n_1+n_2-q-1}.$$

Theorem 6.6: simultaneous intervals with a pooled covariance

Let c_α^2 be as above. With probability $1-\alpha$,

$$\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm c_\alpha \sqrt{\mathbf{a}^\top \left(\tfrac{1}{n_1} + \tfrac{1}{n_2}\right) \mathbf{S}_{\text{pooled}} \mathbf{a}}$$

covers $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ for all $\mathbf{a}$. In particular $\mu_{1j} - \mu_{2j}$ is covered by $(\bar{X}_{1j} - \bar{X}_{2j}) \pm c_\alpha \sqrt{(\frac{1}{n_1} + \frac{1}{n_2})[\mathbf{S}_{\text{pooled}}]_{jj}}$, $j = 1, \ldots, p$.

The proof is as for Theorem 5.12, using $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$.

2.2 Unequal covariances: large samples

Theorem 6.7: simultaneous intervals without a pooled covariance

If $n_1 - p$ and $n_2 - p$ are large, an approximate $100(1-\alpha)\%$ confidence ellipsoid for $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ is

$$[\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)]^\top \left(\tfrac{1}{n_1}\mathbf{S}_1 + \tfrac{1}{n_2}\mathbf{S}_2\right)^{-1} [\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)] \leq \chi^2_p(\alpha),$$

and $\mathbf{a}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ belongs to $\mathbf{a}^\top(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) \pm \sqrt{\chi^2_p(\alpha)} \sqrt{\mathbf{a}^\top(\frac{1}{n_1}\mathbf{S}_1 + \frac{1}{n_2}\mathbf{S}_2)\mathbf{a}}$ simultaneously for all $\mathbf{a}$.

Sketch. (i) $\boldsymbol{\delta}_n := (\bar{\mathbf{x}} - \bar{\mathbf{y}}) - (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \sim N_p(\mathbf{0}, \frac{1}{n_1}\boldsymbol{\Sigma}_1 + \frac{1}{n_2}\boldsymbol{\Sigma}_2)$ (exactly under normality, approximately by the CLT), so $\boldsymbol{\delta}_n^\top (\frac{1}{n_1}\boldsymbol{\Sigma}_1 + \frac{1}{n_2}\boldsymbol{\Sigma}_2)^{-1} \boldsymbol{\delta}_n \sim \chi^2_p$.

(ii) $(n_1 - 1)\mathbf{S}_1 \sim W_p(n_1 - 1, \boldsymbol{\Sigma}_1)$, $(n_2 - 1)\mathbf{S}_2 \sim W_p(n_2 - 1, \boldsymbol{\Sigma}_2)$, and $\mathbf{S}_i \xrightarrow{p} \boldsymbol{\Sigma}_i$.

(iii) For $n_1 = n_2 = n$: $\mathbf{S}_1 + \mathbf{S}_2 \to \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2$, so $(\mathbf{S}_1 + \mathbf{S}_2)^{-1} \to (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}$ and

$$T^2 = n\boldsymbol{\delta}_n^\top (\mathbf{S}_1 + \mathbf{S}_2)^{-1} \boldsymbol{\delta}_n = n\boldsymbol{\delta}_n^\top (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1} \boldsymbol{\delta}_n + n\boldsymbol{\delta}_n^\top [(\mathbf{S}_1 + \mathbf{S}_2)^{-1} - (\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)^{-1}] \boldsymbol{\delta}_n \Rightarrow \chi^2_p,$$

the second term being $O_p(1) \cdot o_p(1)$. ■

3 One-Way MANOVA

3.1 Data and assumptions

Random samples from g populations: population ℓ: $\mathbf{X}_{\ell 1}, \mathbf{X}_{\ell 2}, \ldots, \mathbf{X}_{\ell n_\ell}$, $\ell = 1, \ldots, g$. MANOVA (multivariate analysis of variance) asks whether the mean vectors are equal and, if not, which components differ.

1. $\mathbf{X}_{\ell 1}, \ldots, \mathbf{X}_{\ell n_\ell}$ is a random sample of size n_ℓ from a population with mean $\boldsymbol{\mu}_\ell$; samples from different populations are **independent**.
2. All populations have a **common** covariance matrix $\boldsymbol{\Sigma}$.
3. Each population is **multivariate normal** (can be relaxed by the CLT, Theorem 4.11).

3.2 Review: univariate ANOVA

$X_{\ell j} = \mu + \tau_\ell + e_{\ell j}$, $e_{\ell j}$ independent $N(0,\sigma^2)$, with the constraint $\sum_\ell n_\ell \tau_\ell = 0$ and $H_0 : \tau_1 = \cdots = \tau_g = 0$ ($\mu_\ell = \mu + \tau_\ell$, $\tau_\ell = \mu_\ell - \mu$). The decomposition $x_{\ell j} = \bar{x} + (\bar{x}_\ell - \bar{x}) + (x_{\ell j} - \bar{x}_\ell)$ ($\bar{x}$ estimates μ, $\hat{\tau}_\ell = \bar{x}_\ell - \bar{x}$ estimates τ_ℓ, $x_{\ell j} - \bar{x}_\ell$ estimates $e_{\ell j}$) gives

$$\underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x})^2}_{SS_{\text{cor}}\ \text{(total)}} = \underbrace{\sum_\ell n_\ell (\bar{x}_\ell - \bar{x})^2}_{SS_{\text{tr}}\ \text{(between)}} + \underbrace{\sum_\ell \sum_j (x_{\ell j} - \bar{x}_\ell)^2}_{SS_{\text{res}}\ \text{(within)}},$$

since the cross term is $\sum_\ell (\bar{x}_\ell - \bar{x}) \sum_j (x_{\ell j} - \bar{x}_\ell) = 0$. Equivalently $SS_{\text{obs}} = SS_{\text{mean}} + SS_{\text{tr}} + SS_{\text{res}}$. Reject H_0 if $F = \dfrac{SS_{\text{tr}}/(g-1)}{SS_{\text{res}}/(\sum n_\ell - g)} > F_{g-1, \sum n_\ell - g}(\alpha)$.

3.3 The MANOVA model and the SSP decomposition

Definition 6.8: one-way MANOVA model

$\mathbf{X}_{\ell j} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \mathbf{e}_{\ell j}$, $j = 1, \ldots, n_\ell$, $\ell = 1, \ldots, g$, with $\mathbf{e}_{\ell j}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$. (Indeed, $\boldsymbol{\mu} = \mathrm{E}[\bar{\mathbf{X}}] \iff \sum_\ell \sum_j \mathrm{E}(\mathbf{X}_{\ell j}) = n\boldsymbol{\mu} \iff n\boldsymbol{\mu} + \sum n_\ell \boldsymbol{\tau}_\ell = n\boldsymbol{\mu} \iff \sum n_\ell \boldsymbol{\tau}_\ell = \mathbf{0}$.)

Theorem 6.9: SSP decomposition

From $\mathbf{x}_{\ell j} = \bar{\mathbf{x}} + (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) + (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)$ (overall mean $\hat{\boldsymbol{\mu}}$ + estimated treatment effect $\hat{\boldsymbol{\tau}}_\ell$ + residual $\hat{\mathbf{e}}_{\ell j}$),

$$\underbrace{\sum_{\ell=1}^{g} \sum_{j=1}^{n_\ell} (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top}_{\mathbf{T}=\mathbf{B}+\mathbf{W}} = \underbrace{\sum_{\ell=1}^{g} n_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top}_{\mathbf{B}\text{: between}} + \underbrace{\sum_{\ell=1}^{g} \sum_{j=1}^{n_\ell} (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top}_{\mathbf{W}\text{: within}}.$$

$\mathbf{W} = (n_1 - 1)\mathbf{S}_1 + \cdots + (n_g - 1)\mathbf{S}_g$ generalizes $(n_1 + n_2 - 2)\mathbf{S}_{\text{pooled}}$.

Proof. Expanding, the cross terms are $\sum_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}}) \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top = \mathbf{0}$. ■

Source	Matrix of SSP	d.f.
Treatment	$\mathbf{B} = \sum_\ell n_\ell (\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})(\bar{\mathbf{x}}_\ell - \bar{\mathbf{x}})^\top$	$g - 1$
Residual (error)	$\mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top$	$\sum n_\ell - g$
Total	$\mathbf{B} + \mathbf{W} = \sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top$	$\sum n_\ell - 1$

Remark 6.10: $\mathbf{T} = \mathbf{B} + \mathbf{W}$ as a mixture

Observe $(\mathbf{X}, G)$ with $G \in \{1, \ldots, g\}$, $P(G = i) = \pi_i$, $\mathbf{X} \mid G = i \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$, so $f(\mathbf{x}) = \sum_i f(\mathbf{x} \mid G = i) P(G = i) = \sum_i \phi(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma}) \pi_i$. Then $\mathrm{E}(\mathbf{X}) = \int \mathbf{x} \sum_i \phi(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma}) \pi_i \, d\mathbf{x} = \sum_i \pi_i \boldsymbol{\mu}_i = \boldsymbol{\mu}$ and

$$\mathrm{Var}(\mathbf{X}) = \sum_i \pi_i \mathrm{E}[(\mathbf{x} - \boldsymbol{\mu}_i)(\mathbf{x} - \boldsymbol{\mu}_i)^\top \mid G = i] + \sum_i \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top$$

$$= \boldsymbol{\Sigma} + \sum_i \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top.$$

The first term is the population version of $\mathbf{W}/n$, the second of $\mathbf{B}/n$.

3.4 Wilks' lambda

Definition 6.11: Wilks' test

Reject $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$ if the ratio of generalized variances

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|} = \frac{\left|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)(\mathbf{x}_{\ell j} - \bar{\mathbf{x}}_\ell)^\top\right|}{\left|\sum_\ell \sum_j (\mathbf{x}_{\ell j} - \bar{\mathbf{x}})(\mathbf{x}_{\ell j} - \bar{\mathbf{x}})^\top\right|}$$

is **too small**. Proposed by Wilks, it corresponds to the univariate F test and is related to the likelihood ratio criterion.

Theorem 6.12: Wilks' lambda is the likelihood ratio statistic

With $n = \sum n_\ell$, $\Lambda^{2/n} = \Lambda^*$ and $-2\log\Lambda = n\log\dfrac{|\mathbf{B}+\mathbf{W}|}{|\mathbf{W}|} = n\log\dfrac{|\mathbf{T}|}{|\mathbf{W}|} \dot{\sim} \chi^2_{p(g-1)}$.

Proof. Under H_1: $\hat{\boldsymbol{\mu}}_\ell^{\text{MLE}} = \bar{\mathbf{x}}_\ell$, $\hat{\boldsymbol{\Sigma}}^{\text{MLE}} = \mathbf{S}_p = \dfrac{\sum_\ell (n_\ell - 1)\mathbf{S}_\ell}{n} = \dfrac{\mathbf{W}}{n}$. Under H_0: $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, $\hat{\boldsymbol{\Sigma}} = \dfrac{\mathbf{T}}{n}$. The maximized likelihoods are $(2\pi)^{-np/2}|\hat{\boldsymbol{\Sigma}}|^{-n/2}e^{-np/2}$, so $\Lambda = (|\mathbf{W}|/|\mathbf{T}|)^{n/2}$ and $-2\log(L_0/L_1) = n\log\frac{|\mathbf{T}|}{|\mathbf{W}|}$. Reject H_0 if $n\log\frac{|\mathbf{T}|}{|\mathbf{W}|} > \chi^2_{p(g-1),\alpha}$. ■

Theorem 6.13: distribution of Wilks' lambda for multivariate normal data

No. of variables	No. of groups	Sampling distribution
$p = 1$	$g \geq 2$	$\left(\dfrac{\sum n_\ell - g}{g-1}\right)\left(\dfrac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{g-1,\sum n_\ell - g}$
$p = 2$	$g \geq 2$	$\left(\dfrac{\sum n_\ell - g - 1}{g-1}\right)\left(\dfrac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2(g-1),2(\sum n_\ell - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\dfrac{\sum n_\ell - p - 1}{p}\right)\left(\dfrac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{p,\sum n_\ell - p - 1}$
$p \geq 1$	$g = 3$	$\left(\dfrac{\sum n_\ell - p - 2}{p}\right)\left(\dfrac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2p,2(\sum n_\ell - p - 2)}$

The proof is omitted.

Derivation: the $g = 2$ row — relation to the two-sample T^2

For $g = 2$, $\bar{\mathbf{x}} = (n_1\bar{\mathbf{x}}_1 + n_2\bar{\mathbf{x}}_2)/n$, so $\bar{\mathbf{x}}_1 - \bar{\mathbf{x}} = \frac{n_2}{n}\mathbf{d}$ and $\bar{\mathbf{x}}_2 - \bar{\mathbf{x}} = -\frac{n_1}{n}\mathbf{d}$ with $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$, and

$$\mathbf{B} = n_1\frac{n_2^2}{n^2}\mathbf{d}\mathbf{d}^\top + n_2\frac{n_1^2}{n^2}\mathbf{d}\mathbf{d}^\top = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top.$$

By the determinant lemma $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1 + \frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d})$, and $\mathbf{W} = (n-2)\mathbf{S}_{\text{pooled}}$ gives $\frac{n_1 n_2}{n}\mathbf{d}^\top\mathbf{W}^{-1}\mathbf{d} = \frac{T^2}{n-2}$. Hence

$$\Lambda^* = \left(1 + \frac{T^2}{n-2}\right)^{-1}, \qquad \frac{1-\Lambda^*}{\Lambda^*} = \frac{T^2}{n-2},$$

and Theorem 6.5 ($\frac{n-p-1}{(n-2)p}T^2 \sim F_{p,n-p-1}$) gives the third row. Wilks' test with $g = 2$ is the two-sample T^2 test.

Theorem 6.14: Bartlett's test for Wilks' lambda

If H_0 is true and $\sum n_\ell = n$ is large,

$$-\Big(n-1-\frac{p+g}{2}\Big)\ln\Lambda^* = -\Big(n-1-\frac{p+g}{2}\Big)\ln\Big(\frac{|\mathbf{W}|}{|\mathbf{B}+\mathbf{W}|}\Big) \;\dot\sim\; \chi^2_{p(g-1)},$$

so reject H_0 at level α if the left side exceeds $\chi^2_{p(g-1)}(\alpha)$.

The degrees of freedom $p(g-1)$ are the difference in the number of mean parameters (g mean vectors versus one); the factor $n-1-(p+g)/2$ improves the chi-square approximation.

Remark 6.15: other test statistics

With the eigenvalues $\eta_1 \ge \cdots \ge \eta_s$ of $\mathbf{W}^{-1}\mathbf{B}$ ($s=\min(p,g-1)$),

$$\frac{|\mathbf{T}|}{|\mathbf{W}|} = |\mathbf{I}_p + \mathbf{W}^{-1/2}\mathbf{B}\mathbf{W}^{-1/2}| = \prod_i (1+\eta_i) = \frac{1}{\Lambda^*}, \qquad \text{Lawley–Hotelling } \operatorname{tr}(\mathbf{W}^{-1}\mathbf{B}) = \sum \eta_i,$$

$$\text{Pillai } \operatorname{tr}\{\mathbf{B}(\mathbf{B}+\mathbf{W})^{-1}\} = \sum \frac{\eta_i}{1+\eta_i}, \qquad \text{Roy } \eta_1 = \|\mathbf{W}^{-1}\mathbf{B}\|_2.$$

All summarize "$\mathbf{B}$ is large relative to $\mathbf{W}$". The η_i are also the eigenvalues of Fisher's discriminant analysis (Ch. 10).

Example 6.16: Wisconsin nursing home data

Costs per patient-day: X_1 nursing labour, X_2 dietary labour, X_3 plant operation and maintenance labour, X_4 housekeeping and laundry labour. Ownership: private ($n_1=271$), nonprofit ($n_2=138$), government ($n_3=107$), $\sum n_\ell = 516$. The $\mathbf{S}_\ell$ look compatible and are pooled:

$$\mathbf{W} = \begin{pmatrix} 182.962 & & & \\ 4.408 & 8.200 & & \\ 1.695 & 0.633 & 1.484 & \\ 9.581 & 2.428 & 0.934 & 6.538 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 3.475 & & & \\ 1.111 & 1.225 & & \\ 0.821 & 0.453 & 0.235 & \\ 0.584 & 0.610 & 0.230 & 0.304 \end{pmatrix}$$

(symmetric; lower triangles shown), $\bar{\mathbf{x}} = (2.136, 0.519, 0.102, 0.380)^\top$. With $\Lambda^* = 0.7714$, $p=4$, $g=3$:

$$\Big(\frac{\sum n_\ell - p - 2}{p}\Big)\Big(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\Big) = \Big(\frac{516-4-2}{4}\Big)\Big(\frac{1-\sqrt{.7714}}{\sqrt{.7714}}\Big) = 17.67 > F_{8,1020}(.01) \approx 2.51.$$

Reject H_0 at the 1% level: average costs differ by type of ownership. (Bartlett: $-(516-1-3.5)\ln 0.7714 = 132.8 > \chi^2_8(.01) = 20.09$.)

4 Simultaneous Intervals for Treatment Effects

Let τ_{kj} be the jth component of $\boldsymbol{\tau}_k$. Since $\hat{\boldsymbol{\tau}}_k = \bar{\mathbf{X}}_k - \bar{\mathbf{X}}$, $\hat\tau_{kj} - \hat\tau_{\ell j} = \bar X_{kj} - \bar X_{\ell j}$ and a two-sample t-based interval is valid with an appropriately corrected α:

$$\operatorname{Var}(\hat\tau_{kj} - \hat\tau_{\ell j}) = \Big(\frac{1}{n_k}+\frac{1}{n_\ell}\Big)\sigma_{jj}, \qquad \widehat{\operatorname{Var}}(\bar X_{kj} - \bar X_{\ell j}) = \Big(\frac{1}{n_k}+\frac{1}{n_\ell}\Big)\frac{\mathbf{W}_{jj}}{n-g}.$$

Apportioning the error rate: for p variables and $g(g-1)/2$ pairwise differences there are $m = p\,g(g-1)/2$ statements, so each interval uses $t_{n-g}(\alpha/2m)$.

Theorem 6.17: simultaneous intervals for differences in treatment effects

Let $n = \sum_k n_k$. With confidence at least $1-\alpha$,

$$\tau_{kj} - \tau_{\ell j} \quad \text{belongs to} \quad \bar{x}_{kj} - \bar{x}_{\ell j} \pm t_{n-g}\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{\mathbf{W}_{jj}}{n-g}\Big(\frac{1}{n_k}+\frac{1}{n_\ell}\Big)}$$

for all components $j = 1, \ldots, p$ and all differences $\ell < k = 1, \ldots, g$.

Example 6.18: nursing home cost comparison

Compare plant operation and maintenance (X_3) between private and government homes, i.e. estimate $\tau_{13} - \tau_{33}$: $\hat{\boldsymbol{\tau}}_1 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}) = (-.070, -.039, -.020, -.020)^\top$, $\hat{\boldsymbol{\tau}}_3 = (\bar{\mathbf{x}}_3 - \bar{\mathbf{x}}) = (.137, .002, .023, .003)^\top$, so $\hat{\tau}_{13} - \hat{\tau}_{33} = -.020 - .023 = -.043$. With $n = 271+138+107 = 516$: $\sqrt{(\frac{1}{271}+\frac{1}{107})\frac{1.484}{516-3}} = 0.00614$; for $p = 4$, $g = 3$ the 95% interval uses $t_{513}(.05/4(3)2) = t_{513}(.00208) \approx 2.87$. Hence $\tau_{13} - \tau_{33} \in -0.043 \pm 2.87 \times 0.00614 = (-0.061, -0.025)$: government homes have higher plant operation costs.

5 Two-Way MANOVA

5.1 Review: univariate two-way ANOVA

Factor 1 with g levels, factor 2 with b levels, n independent observations at each of the gb combinations; $X_{\ell kr}$ is the rth observation at level ℓ of factor 1 and level k of factor 2:

$$X_{\ell kr} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\tau_\ell}_{\text{factor 1 effect}} + \underbrace{\beta_k}_{\text{factor 2 effect}} + \underbrace{\gamma_{\ell k}}_{\text{interaction}} + \underbrace{e_{\ell kr}}_{\text{normal error}},$$

$\sum_\ell \tau_\ell = \sum_k \beta_k = \sum_\ell \gamma_{\ell k} = \sum_k \gamma_{\ell k} = 0$, and $SS_{\text{cor}} = SS_{\text{fac1}} + SS_{\text{fac2}} + SS_{\text{int}} + SS_{\text{res}}$:

Source	Sum of squares	d.f.
Factor 1	$SS_{\text{fac1}} = \sum_\ell bn(\bar{x}_{\ell\cdot} - \bar{x})^2$	$g-1$
Factor 2	$SS_{\text{fac2}} = \sum_k gn(\bar{x}_{\cdot k} - \bar{x})^2$	$b-1$
Interaction	$SS_{\text{int}} = \sum_\ell \sum_k n(\bar{x}_{\ell k} - \bar{x}_{\ell\cdot} - \bar{x}_{\cdot k} + \bar{x})^2$	$(g-1)(b-1)$
Residual	$SS_{\text{res}} = \sum_\ell \sum_k \sum_r (x_{\ell kr} - \bar{x}_{\ell k})^2$	$gb(n-1)$
Total	$SS_{\text{cor}} = \sum\sum\sum (x_{\ell kr} - \bar{x})^2$	$gbn-1$

F-ratios: $F_{\text{fac1}} = \frac{SS_{\text{fac1}}/(g-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{fac2}} = \frac{SS_{\text{fac2}}/(b-1)}{SS_{\text{res}}/(gb(n-1))}$, $F_{\text{int}} = \frac{SS_{\text{int}}/((g-1)(b-1))}{SS_{\text{res}}/(gb(n-1))}$.

5.2 Multivariate two-way fixed-effects model with interaction

Definition 6.19: two-way MANOVA

$\mathbf{X}_{\ell kr} = \boldsymbol{\mu} + \boldsymbol{\tau}_\ell + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{\ell k} + \mathbf{e}_{\ell kr}$ with $\mathbf{e}_{\ell kr}$ independent $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $\sum_\ell \boldsymbol{\tau}_\ell = \sum_k \boldsymbol{\beta}_k = \sum_\ell \boldsymbol{\gamma}_{\ell k} = \sum_k \boldsymbol{\gamma}_{\ell k} = \mathbf{0}$; all vectors are $p \times 1$.

The SSP decomposition $\text{SSP}_{\text{cor}} = \text{SSP}_{\text{fac1}} + \text{SSP}_{\text{fac2}} + \text{SSP}_{\text{int}} + \text{SSP}_{\text{res}}$ replaces each $(\cdot)^2$ by $(\cdot)(\cdot)^\top$, e.g. $\text{SSP}_{\text{int}} = \sum_\ell \sum_k n(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})(\bar{\mathbf{x}}_{\ell k} - \bar{\mathbf{x}}_{\ell\cdot} - \bar{\mathbf{x}}_{\cdot k} + \bar{\mathbf{x}})^\top$, with the same degrees of freedom.

Theorem 6.20: two-way MANOVA tests (Bartlett-corrected)

Interaction, $H_0 : \boldsymbol{\gamma}_{11} = \boldsymbol{\gamma}_{12} = \cdots = \boldsymbol{\gamma}_{gb} = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{int}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)(b-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)(b-1)p}(\alpha).$$

Main effects of factor 1, $H_0 : \boldsymbol{\tau}_1 = \cdots = \boldsymbol{\tau}_g = \mathbf{0}$: with $\Lambda^* = \dfrac{|\mathrm{SSP}_{\mathrm{res}}|}{|\mathrm{SSP}_{\mathrm{fac1}} + \mathrm{SSP}_{\mathrm{res}}|}$, reject if

$$-\Big[gb(n-1) - \frac{p+1-(g-1)}{2}\Big] \ln \Lambda^* > \chi^2_{(g-1)p}(\alpha).$$

Factor 2 is the same with $g-1$ replaced by $b-1$. Test the interaction first; if present, interpret main effects with care.

Theorem 6.21: simultaneous intervals in two-way MANOVA

With $\nu = gb(n-1)$ and E_{jj} the jth diagonal element of $\mathbf{E} = \mathrm{SSP}_{\mathrm{res}}$,

$$\tau_{\ell j} - \tau_{mj} \quad \text{belongs to} \quad (\bar{x}_{\ell\cdot j} - \bar{x}_{m\cdot j}) \pm t_\nu\Big(\frac{\alpha}{pg(g-1)}\Big)\sqrt{\frac{E_{jj}}{\nu}\cdot\frac{2}{bn}}.$$